

A micromorphic constitutive framework for (non)linear softening modeling

Rafael Abreu^{1,*}

¹ *Institut de Physique du Globe de Paris, CNRS, Université de Paris, Paris, France*

* corresponding author: rabreu@ipgp.fr

Abstract

Nonlinear shear softening in soils and rocks is commonly described using empirical modulus-reduction laws. In this work, we investigate nonlinear softening within the Eringen–Mindlin micromorphic framework by interpreting its additional constitutive parameters in terms of coupling, internal relaxation, and micro-inertia. The micromorphic continuum is viewed as a dynamically enriched effective medium in which the macroscopic deformation interacts with an internal microstructural deformation field. This interaction naturally produces frequency-dependent effective elastic properties and provides a physical connection between macroscopic wave propagation and unresolved microstructural dynamics.

We show that nonlinear shear softening can be described by a progressive reduction of the coupling between macroscopic and internal deformation. As this coupling weakens with increasing shear strain, the effective shear modulus decreases. In the low-frequency limit, the resulting constitutive relation takes a hyperbolic form closely related to the classical Hardin–Drnevich law. The model is validated against laboratory measurements of shear-modulus reduction in sand and reproduces the observed nonlinear behavior with only a small number of physically interpretable parameters.

The proposed framework provides a new description of nonlinear softening and frequency-dependent behavior without requiring explicit knowledge of the underlying microstructural geometry. Classical empirical modulus-reduction curves emerge as a macroscopic consequence of the interaction between macroscopic and internal deformation.

Keywords: micromorphic theory, nonlinear shear softening, internal relaxation, dynamic homogenization, seismic wave propagation, geomaterials

1 Introduction

Micro-continuum theories (Mindlin, 1964; Eringen, 1999) provide a natural framework to extend classical linear elasticity by incorporating additional internal degrees of freedom. The micro-continuum field theories cover a wide range of three main different theories: micropolar, microstretch and micromorphic media (Eringen, 1999). The theory of Cosserat/micropolar media (Cosserat and Cosserat, 1909), describes a continuum material with additional free rotational degrees of freedom at each location (Eringen and Kafadar, 1976; Nowacki, 1986; Jeong and Neff, 2010; Neff and Jeong, 2009; Kulesh et al., 2009; Grekova et al., 2009; Grekova, 2012a,b, 2016). Each particle inside the material can independently rotate. The theory of microstretch media describes elastic materials with rotational and stretching free degrees of freedom (Eringen, 1999). Each particle inside the material can only independently rotate and have stretching (breathing) motions. The theory of micromorphic media describes a continuum that allows any kind of affine micro-distortion. While these models promise a powerful description of media with microstructure, their practical application is often limited by the large number of constitutive parameters and the difficulty of assigning them a clear physical meaning.

The application of micro continuum field theories in the Earth sciences is not new (e.g. Teisseyre, 1973; Teisseyre et al., 2006; Nagahama and Teisseyre, 2000a,b; Teisseyre et al., 2008; Teisseyre, 2008, 2011). Pioneering work by Teisseyre (1973, 1974) introduced a simplified general micromorphic model to describe earthquake processes, based on a symmetric microstrain formulation, as a particular case of the general micromorphic theory proposed by Eringen and Claus Jr (1970). However, due to its symmetric formulation, this early model is unable to represent micro-rotational deformations. This limitation motivates the use of more general formulations, such as the relaxed micromorphic model, in which both symmetric and antisymmetric microstructural effects are effectively modeled.

A different model has been introduced by Teisseyre and Górski (2009) in order to describe the earthquake source processes. This model relies on the consideration of a different perspective of motions and deformations in an asymmetric continuum, including simple motions (translation and rotation, named spin) and simple deformations. It is based on the constitutive relation introduced by Shimbo (1975) joining the rotations with the antisymmetric stresses (Teisseyre and Górski, 2007)¹.

Despite these advances in the Earth sciences, the role of microstructure in seismic wave propagation remains still poorly understood. In particular, classical approaches to account for microstructural effects in seismology are typically based on homogenization theory, in which small-scale heterogeneities are averaged out to obtain effective macroscopic properties (e.g. Capdeville et al., 2010, 2013, 2015; Capdeville and Métivier, 2018; Hedjazian et al., 2021). While these methods provide a rigorous link between microscale structure and large-scale behavior under conditions of strong scale separation, they lead to effective models in which the microstructure no longer appears explicitly. As a consequence, the resulting description is purely macroscopic and does not retain internal degrees of freedom that could participate dynamically in wave propagation. As we will see, this limitation becomes particularly relevant when the microstructure influences wave propagation behavior within an observable frequency range. In these situations, seismic observables such as velocity dispersion, modulus reduction, and frequency-dependent responses cannot be directly linked to underlying physical mechanisms, since the microstructure has been entirely eliminated from the mathematical description. As a result, different physical processes may lead to indistinguishable macroscopic behavior, limiting the interpretability of seismic observations. In such cases, an alternative modeling strategy is required, in which microstructural dynamics are not eliminated but incorporated in the mathematical description of the wave dynamics (Eringen, 1999).

We investigate linear elastic wave propagation within the Eringen-Mindlin micromorphic framework, with a particular focus on a new physical interpretation of its constitutive parameters and realistic Earth sciences applications. We show that the required micromorphic model parameters characterize the fundamental physics that governs micro and macro scales, providing a direct fundamental physical description of the underlying dynamical mechanisms. Within this perspective, macroscopic wave behavior naturally emerges from the interaction between elastic response and internal relaxation mechanisms.

¹the independent rotation field and related constitutive law joining the antisymmetric stresses and rotations, e.g., rotations related to grains (the points of continuum), have been considered by Shimbo (1975) in his considerations on the friction and fracturing processes. In this way, there arose an idea to construct the theory of a continuum with asymmetric force stresses.

This paper is now organized as follows. In Section 2 we present the micromorphic theoretical framework within a clear notation and with a particular emphasis to the physical interpretation of the model parameters. In Section 3, we validate the micromorphic model against experimental and geophysical observations in experiments of strain-dependent shear softening. Section 4 provides a broader interpretation of the results. We further discuss connections with homogenization theory. Finally, conclusions are drawn in Section 5.

2 Theoretical foundations

The equation of motion of any continua can be found, in a simple way, by using the Lagrangian formalism also called the variational principle of least action and/or the principle of stationary action. The Lagrangian formalism aims to find the extremal functional(s) that make the functional S obtain a minimum value (principle of stationary action $\delta S = 0$). The functional S can be written in general as follows

$$S = \int_0^t \int_{\Omega} \mathcal{L} \, dx \, dt = \int_0^t \int_{\Omega} (K - \mathcal{E}) \, dx \, dt, \quad (1)$$

where Ω denotes the volume of the continuum, $\mathcal{L}$ is the Lagrangian defined as the difference between kinetic energy K and potential energy $\mathcal{E}$. The functional(s) that minimizes S is simply found by setting $\delta S = 0$.

It is thus straightforward to find equations of motion in continuum media including micro-structure (or not), by defining the appropriate kinetic K and potential (internal elastic free) $\mathcal{E}$ energies. The potential energy $\mathcal{E}$ is of main importance, since depending on how we define $\mathcal{E}$, the definition of the kinetic energy is straightforward in a first attempt.

2.1 Internal free energy

In conventional linear elastic media, the potential energy $\mathcal{E}$ is defined as follows (see [Slawinski \(2015\)](#) for further details)

$$\mathcal{E}(\varepsilon) = \frac{1}{2} \varepsilon_{ij} \mathbb{C}_{ijkl} \varepsilon_{kl}, \quad \text{with} \quad \varepsilon_{ij} = \partial_j u_i + \partial_i u_j, \quad (2)$$

where $u : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ denotes the macro-displacement and $\mathbb{C}_{ijkl}$ is a fourth-order tensor of elastic constants. In the rest of this paper we will denote the fourth-order tensor of elastic constants related to the linear elasticity as

$$(\mathbb{C}_{\text{macro}})_{ijkl} = \mathbb{C}_{ijkl}. \quad (3)$$

Note that the strain tensor ε_{ij} is symmetric ($\varepsilon_{ij} = \varepsilon_{ji}$) and it does not involve any other physical quantities than the displacement u . In contrast, the general internal elastic free energy for the Eringen-Mindlin micromorphic continuum can be written as follows

$$\mathcal{E}(\varepsilon, \epsilon, e, \gamma) = \underbrace{\frac{1}{2} \varepsilon_{ij} \mathbb{A}_{ijkl} \epsilon_{kl}}_{\text{elastic relative free energy}} + \underbrace{\frac{1}{2} e_{ij} \mathbb{B}_{ijkl} e_{kl}}_{\text{relaxation free energy}} + \underbrace{\frac{1}{2} \gamma_{klm} \mathbb{L}_{klmnpq} \gamma_{npq}}_{\text{curvature energy}}, \quad (4)$$

where $\mathbb{A}, \mathbb{B}$, are fourth-order tensors and $\mathbb{L}$ a six-order tensor of curvature elastic constants ([Eringen, 1999](#)). The strain measures $(\varepsilon, \epsilon, e, \gamma)$ are defined by the following expressions

$$\epsilon_{kl} = \partial_k u_l - P_{kl}, \quad e_{kl} = P_{kl}, \quad \gamma_{klm} = \partial_m P_{kl}. \quad (5)$$

In a symbolic formulation we write

$$\begin{aligned} \mathcal{E}(\varepsilon, \epsilon, e, \gamma) &= \langle (\nabla u - P), \mathbb{A}(\nabla u - P) \rangle + \langle P, \mathbb{B}P \rangle + \langle \nabla P, \mathbb{L} \nabla P \rangle, \\ &= \langle \text{sym}(\nabla u - P), \mathbb{C}_{\text{rel}} \text{sym}(\nabla u - P) \rangle + \langle \text{skew}(\nabla u - P), \mathbb{C}_{\text{rot}} \text{skew}(\nabla u - P) \rangle \\ &\quad + \langle \text{sym} P, \mathbb{C}_{\text{relax}} \text{sym} P \rangle + \langle \text{skew} P, \mathbb{C}_{\text{spin}} \text{skew} P \rangle + \langle \nabla P, \mathbb{L} \nabla P \rangle, \end{aligned} \quad (6)$$

where $u : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ denotes the usual macro-displacement and $P : \mathbb{R}^3 \rightarrow \mathbb{R}^{3 \times 3}$ denotes the micro-distortion second-order tensor allowing for the extra degrees of freedom: the skew-symmetric part—skew P —represents independent rotational degrees of freedom, the spherical part— $\frac{1}{3} \text{tr} (P)$ —represents possible breathing motions and the deviatoric symmetric part—dev sym P —involves independent stretch (volume preserving) motions. The terms $\mathbb{C}_{\text{rel}}, \mathbb{C}_{\text{rot}}, \mathbb{C}_{\text{relax}}, \mathbb{C}_{\text{spin}}$ are fourth-order tensors of elastic constants.

One can note that the internal elastic energy for the Eringen-Mindlin micromorphic model (eq. (4)) is much more complex compared to the linear elastic case (eq. (2)). The main and most important difference is that the strain at the macro-structural level given by— ∇u —is different from the strain at the local-structural level— P .

Formally, the resulting stress measures are then given by the following expressions

$$\begin{aligned}\sigma_{ij} &= \partial_e \mathcal{E} = \mathbb{A}_{ijkl} \epsilon_{kl} = (\mathbb{C}_{\text{rel}})_{ijkl} \text{sym } \epsilon_{kl} + (\mathbb{C}_{\text{rot}})_{ijkl} \text{skew } \epsilon_{kl}, \\ s_{ij} &= \partial_e \mathcal{E} = \mathbb{B}_{ijkl} e_{kl} = (\mathbb{C}_{\text{relax}})_{ijkl} \text{sym } e_{kl} + (\mathbb{C}_{\text{spin}})_{ijkl} \text{skew } e_{kl}, \\ m_{klm} &= \partial_\gamma \mathcal{E} = \mathbb{L}_{klmnpq} \gamma_{npq},\end{aligned}\tag{7}$$

where σ is called the force-stress tensor, m is called stress-moment tensor and s , we call the relaxation-stress tensor, and the supper indices *sym* and *skew* refer to the symmetric and anti-symmetric parts respectively. We can explicitly write the general resulting stress measures as follows

$$\begin{aligned}\sigma &= \mathbb{C}_{\text{rel}} \text{sym} (\nabla u - P) + \mathbb{C}_{\text{rot}} \text{skew} (\nabla u - P), \\ s &= \mathbb{C}_{\text{relax}} \text{sym} P + \mathbb{C}_{\text{spin}} \text{skew} P, \\ m &= L_c^2 \mathbb{C}_{\text{macro}} \nabla P,\end{aligned}\tag{8}$$

where, for simplicity, we have considered $\mathbb{L} = L_c^2 \mathbb{C}_{\text{macro}}$. The isotropic stress measures corresponding to eqs. (8) are given by the following expressions

$$\begin{aligned}\sigma &= 2\mu_{\text{rel}} \text{sym} (\nabla u - P) + 2\mu_{\text{rot}} \text{skew} (\nabla u - P) + \lambda_{\text{rel}} \text{tr} (\nabla u - P) \cdot \mathbb{1}, \\ s &= 2\mu_{\text{relax}} \text{sym} P + 2\mu_{\text{spin}} \text{skew} P + \lambda_{\text{relax}} \text{tr} (P) \cdot \mathbb{1}, \\ m &= L_c^2 \mu_{\text{macro}} \nabla P,\end{aligned}\tag{9}$$

where the parameter μ_{macro} is having the role of the classical Lamé shear modulus. A new set of elastic constants $\lambda_{\text{rel}}, \lambda_{\text{relax}}, \mu_{\text{rel}}, \mu_{\text{relax}}, \mu_{\text{spin}} \geq 0$ are introduced and need to be determined. The elastic constant μ_{rot} is called the *Cosserat couple modulus*. Finally L_c is a characteristic length related to the problem in study.

2.2 Kinetic energy and equations of motion

Having defined the elastic free energy eq. (4), it is straightforward to formally define the corresponding kinetic energy as follows

$$K = \frac{1}{2} \rho \partial_t u \partial_t u + \frac{1}{2} \eta \partial_t P \partial_t P,\tag{10}$$

where ρ is the macroscopic scalar density, η is the micro-inertia density tensor of the micro-distortion (in the isotropic case simply a scalar). Solving the stationary condition problem $\delta S = 0$, with S defined by eq. (1), we can find the general, second-order coupled, micromorphic equations of motion (Eringen, 1999)

$$\begin{aligned}\rho \partial_t^2 u &= \text{Div} [\mathbb{C}_{\text{rel}} \text{sym} (\nabla u - P) + \mathbb{C}_{\text{rot}} \text{skew} (\nabla u - P)], \\ \eta \partial_t^2 P &= \text{Div} [L_c^2 \mathbb{C}_{\text{macro}} \nabla P] + [\mathbb{C}_{\text{rel}} \text{sym} (\nabla u - P) + \mathbb{C}_{\text{rot}} \text{skew} (\nabla u - P)] \\ &\quad - \mathbb{C}_{\text{relax}} \text{sym} P - \mathbb{C}_{\text{spin}} \text{skew} P,\end{aligned}\tag{11}$$

with the free surface boundary conditions given by

$$\sigma \cdot \hat{n} = 0 \quad \text{on} \quad \partial \Omega, \quad m \cdot \hat{n} = 0 \quad \text{on} \quad \partial \Omega,\tag{12}$$

where $\hat{n}$ refers to the direction normal to the surface $\partial \Omega$.

2.3 The relaxed micromorphic model

The relaxed micromorphic model introduced in [Neff et al. \(2014\)](#) attempts to reduce the number of parameters required by the original Eringen/M Mindlin model and to provide clarity in the application of micromorphic theory. Within the relaxed micromorphic framework, the number of independent parameters is significantly reduced, while retaining the ability to describe nonlocal effects, dispersion, and scale-dependent behavior. It has been recently applied in the engineering field for the design of seismic metamaterials (e.g. [Madeo et al., 2015, 2016b,a](#); [Hermann et al., 2024](#); [Demetriou et al., 2025](#); [Madeo et al., 2017](#); [Hermann et al., 2026](#); [Barbagallo et al., 2019](#); [Rizzi et al., 2024](#); [d'Agostino et al., 2020](#); [Voss et al., 2023](#); [Ramirez et al., 2024](#); [Madeo et al., 2016b,a](#); [Neff et al., 2020](#)).

The general elastic energy is given by the following expression

$$\begin{aligned}\mathcal{E}(\varepsilon, \epsilon, e, \gamma) &= \langle (\nabla u - P), \mathbb{A}(\nabla u - P) \rangle + \langle \text{sym } P, \mathbb{B} \text{sym } P \rangle + \langle \text{Curl } P, \mathbb{L} \text{Curl } P \rangle, \\ &= \langle \text{sym } (\nabla u - P), \mathbb{C}_{\text{rel}} \text{sym } (\nabla u - P) \rangle + \langle \text{skew } (\nabla u - P), \mathbb{C}_{\text{rot}} \text{skew } (\nabla u - P) \rangle \\ &\quad + \langle \text{sym } P, \mathbb{C}_{\text{relax}} \text{sym } P \rangle + \langle \text{Curl } P, \mathbb{L} \text{Curl } P \rangle,\end{aligned}\quad (13)$$

The main conceptual difference with the original Eringen–Mindlin model lies in relaxing the requirement that the full gradient of the micro-deformation, ∇P , be energetically penalized. The underlying idea is that not all components of ∇P are required to capture the mechanically relevant nonlocal effects. Therefore, the curvature energy is formulated in terms of reduced differential operators acting on P , such as $\text{Curl } P$, $\text{Div } P$, or any other combination. Thus the relaxed micromorphic model corresponds to any of those particular choices (e.g. [Madeo et al., 2016a](#)).

However, it is important to emphasize that the decomposition of the relative deformation into its symmetric and skew-symmetric parts,

$$\nabla u - P = \text{sym } (\nabla u - P) + \text{skew } (\nabla u - P), \quad (14)$$

is not unique to the relaxed micromorphic model. Rather, the same decomposition follows naturally from the general Eringen–Mindlin formulation and is therefore equally valid within the original Eringen–Mindlin framework, independently of the particular choice of curvature energy.

2.4 Main contributions of this work

For the purposes of the present work, we adopt the constitutive framework of the Eringen–Mindlin micromorphic theory while expressing the relative deformation energy through its decomposition into symmetric and skew-symmetric parts. When the curvature energy is neglected, i.e., $\nabla P = \text{Curl } P = 0$, the remaining local elastic energy is identical in both the Eringen–Mindlin and relaxed micromorphic formulations.

The objective of the present work is not to propose a new micromorphic constitutive model, but rather to provide a new physical interpretation of the classical Eringen–Mindlin formulations and to validate it against experimental observations.

While the governing equations of Eringen–Mindlin formulation remain unchanged, the constitutive tensors are reinterpreted from a new mechanical perspective. To the best of the author’s knowledge, this physical interpretation has not been presented previously.

The remainder of this work we adopt the notation $\mathbb{C}_{\text{rel}}$ and $\mathbb{C}_{\text{rot}}$ to denote the energetic contributions associated with the symmetric and skew-symmetric parts of the relative deformation tensor $(\nabla u - P)$. This notation is used exclusively to facilitate the physical interpretation developed in the following sections.

In particular, the tensors $\mathbb{C}_{\text{rel}}$, $\mathbb{C}_{\text{rot}}$, $\mathbb{C}_{\text{relax}}$, and $\mathbb{C}_{\text{spin}}$ are interpreted as energetic resistances associated with distinct deformation mechanisms rather than merely as constitutive coefficients. As we will see, within the introduced framework, $\mathbb{C}_{\text{rel}}$ and $\mathbb{C}_{\text{rot}}$ govern the resistance to the relative deformation between the macroscopic and microstructural fields, whereas $\mathbb{C}_{\text{relax}}$ and $\mathbb{C}_{\text{spin}}$ quantify the energetic resistance to the activation of internal deformation modes.

This new interpretation naturally leads to a dynamic homogenization viewpoint, in which the effective constitutive response emerges from the interaction between the macroscopic deformation and an local/microstructural

deformation. As a consequence, the constitutive parameters acquire a direct mechanical interpretation that, to the best of the author's knowledge, is absent from the original Eringen–Mindlin and relaxed micromorphic formulations.

2.5 Normal mode solutions

Considering $L_c = 0$, we can write the general equations of motion (eq. (11)) as follows

$$\begin{aligned}\rho \partial_t^2 u &= \text{Div} [\mathbb{C}_{\text{rel}} \text{sym} (\nabla u - P) + \mathbb{C}_{\text{rot}} \text{skew} (\nabla u - P)], \\ \eta \partial_t^2 P &= [\mathbb{C}_{\text{rel}} \text{sym} (\nabla u - P) + \mathbb{C}_{\text{rot}} \text{skew} (\nabla u - P)] - \mathbb{C}_{\text{relax}} \text{sym} P - \mathbb{C}_{\text{spin}} \text{skew} P.\end{aligned}\quad (15)$$

We next assume normal modes solutions of the form

$$u(x, t) = \varphi(x) e^{i\omega t}, \quad P(x, t) = \Psi(x) e^{i\omega t}, \quad (16)$$

where $\varphi : \Omega \rightarrow \mathbb{R}^3$ and $\Psi : \Omega \rightarrow \mathbb{R}^{3 \times 3}$ are spatial eigenfunctions defined on the domain Ω , and ω denotes the associated eigenfrequency.

Substitution of the normal mode solutions (eq. (16)) into the governing equations of motion (eq. (11)), leads to a coupled eigenvalue problem for (φ, Ψ) and ω^2 . From the second equation in eq. (15), the independent deformation field Ψ is linearly related to the macroscopic deformation gradient $\nabla \varphi$ as follows

$$\begin{aligned}\left[(\mathbb{C}_{\text{rel}} + \mathbb{C}_{\text{relax}}) - \eta \omega^2 \mathbb{1} \right] \text{sym} \Psi &= \mathbb{C}_{\text{rel}} \text{sym} (\nabla \varphi), \\ \left[(\mathbb{C}_{\text{rot}} + \mathbb{C}_{\text{spin}}) - \eta \omega^2 \mathbb{1} \right] \text{skew} \Psi &= \mathbb{C}_{\text{rot}} \text{skew} (\nabla \varphi),\end{aligned}\quad (17)$$

where $\mathbb{1}$ denotes the fourth-order identity tensor. If we insert eq. (17) into the first equation of eq. (15), and consider the anti-symmetric part to be zero, we obtain the effective condition

$$\mathbb{C}_{\text{macro}} = \mathbb{C}_{\text{rel}} - \mathbb{C}_{\text{rel}} \left[\mathbb{C}_{\text{e}} + \mathbb{C}_{\text{relax}} - \eta \omega^2 \mathbb{1} \right]^{-1} \mathbb{C}_{\text{rel}}. \quad (18)$$

The effective elasticity tensor $\mathbb{C}_{\text{macro}}$ (defined in eq. (3)) is frequency dependent and it possesses a pole whenever

$$\mathbb{C}_{\text{rel}} + \mathbb{C}_{\text{relax}} - \eta \omega^2 \mathbb{1} = 0, \quad (19)$$

for which the expression is no longer well defined. At these frequencies the constitutive model is no longer well defined, and dissipative effects must be incorporated.

Equation (18) simply shows that we are able to map the micromorphic model to an effective linear elastic model with frequency dependent parameters (see Fig. 1). We can observe that the effective frequency dependent macroscopic response ($\mathbb{C}_{\text{macro}} = \mathbb{C}_{\text{macro}}(\omega)$) emerges from the interplay between the elastic stiffness $\mathbb{C}_{\text{rel}}$, the relaxation stiffness $\mathbb{C}_{\text{relax}}$, and the dynamic contributions associated with micro-inertia term $\eta \omega^2$. As a result, the effective elastic response becomes frequency dependent, that can be understood as emergent form the general microstructural internal dynamics.

In the limiting case where the micro-inertia is negligible, i.e., $\eta \rightarrow 0$, the dynamic contributions vanish and the static consistency condition is recovered

$$\mathbb{C}_{\text{macro}} = \mathbb{C}_{\text{rel}} \mathbb{C}_{\text{relax}} (\mathbb{C}_{\text{rel}} + \mathbb{C}_{\text{relax}})^{-1}. \quad (20)$$

Using the skew-symmetric part of the equation of motion (eq. (15)), and eq. (17) we can write

$$\mathbb{C}_{\text{rot}} \text{skew} (\nabla \varphi - \Psi) = (\mathbb{C}_{\text{spin}} - \eta \omega^2 \mathbb{1}) \text{skew} \Psi. \quad (21)$$

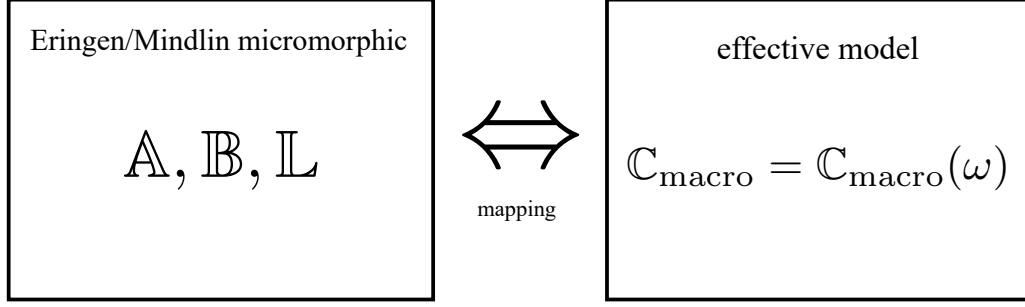


Figure 1: Micromorphic mapping to an linear elastic model with frequency dependent elastic parameters.

Since the effective macroscopic elasticity tensor $\mathbb{C}_{\text{macro}}$ is symmetric, the skew-symmetric contribution must vanish. This is satisfied either when skew $\Psi = 0$ or when

$$\mathbb{C}_{\text{spin}} - \eta \omega^2 \mathbb{1} = 0. \quad (22)$$

This condition represents a dynamic balance between the rotational elastic stiffness $\mathbb{C}_{\text{spin}}$ and the micro-inertia η . Under this condition, the skew-symmetric part of the relative deformation no longer contributes to the macroscopic stress, i.e.,

$$\mathbb{C}_{\text{rot}} \text{skew} (\nabla \varphi - \Psi) = 0, \quad (23)$$

so that the effective macroscopic constitutive response is purely symmetric and can therefore be completely described by the effective elasticity tensor $\mathbb{C}_{\text{macro}}$.

2.6 Transverse 1D waves

Assuming one-dimensional transverse wave propagation, the problem reduces to a single active deformation mode. Following the same procedure as before, we can write

$$\mu_{\text{macro}} = \mu_{\text{rel}} \frac{\mu_{\text{relax}} - \eta \omega^2}{\mu_{\text{rel}} + \mu_{\text{relax}} - \eta \omega^2}, \quad (24)$$

with the two limits

$$\lim_{\omega \rightarrow 0} \mu_{\text{macro}} = \frac{\mu_{\text{rel}} \mu_{\text{relax}}}{\mu_{\text{rel}} + \mu_{\text{relax}}}, \quad \lim_{\omega \rightarrow \infty} \mu_{\text{macro}} = \mu_{\text{rel}}, \quad (25)$$

This means that at very high frequencies, the microstructure cannot follow the fast oscillations, so the response is dominated by the effective shear stiffness μ_{rel} .

The transition between these two limiting regimes is governed by the internal inertia of the microstructure. In particular, the effective modulus exhibits a zero at

$$\omega_z = \sqrt{\frac{\mu_{\text{micro}}}{\eta}}, \quad (26)$$

and a pole at

$$\omega_p = \sqrt{\frac{\mu_{\text{rel}} + \mu_{\text{micro}}}{\eta}}. \quad (27)$$

The vanishing of $\mu_{\text{macro}}(\omega)$ at ω_z corresponds to a complete cancellation of the effective shear stiffness, whereas the pole at ω_p represents the natural resonance frequency of the internal microstructural degree of freedom. This means that, the effective modulus cannot be understood as an interpolation between μ_{macro} and μ_{rel} , but exhibits a resonant transition controlled by the micro-inertia η . In practical materials this singular behavior is regularized by dissipation or viscosity, yielding a finite resonant peak instead of an ideal mathematical pole.

2.7 Longitudinal 1D waves

In 1D and for longitudinal waves, we get

$$C_{\text{macro}} = C_e \frac{C_{\text{relax}} - \eta \omega^2}{C_e + C_{\text{relax}} - \eta \omega^2}, \quad (28)$$

where for keeping generality, we have denoted $C_{\text{macro}} = 2\mu_{\text{macro}} + \lambda_{\text{macro}}$, $C_e = 2\mu_e + \lambda_e$, $C_{\text{relax}} = 2\mu_{\text{relax}} + \lambda_{\text{relax}}$ denote the scalar projections of the corresponding fourth-order tensors onto the considered longitudinal mode. The frequency-dependent longitudinal modulus exhibits a zero at

$$\omega_z = \sqrt{\frac{C_{\text{relax}}}{\eta}}, \quad (29)$$

and a pole at

$$\omega_p = \sqrt{\frac{C_e + C_{\text{relax}}}{\eta}}. \quad (30)$$

Note that both the effective longitudinal and transverse moduli possess the same frequency-dependent rational structure, characterized by a zero and a resonance pole. This behavior is a direct consequence of the dynamic condensation of the internal microdeformation field and is therefore independent of the particular deformation mode.

2.8 Physical interpretation of elastic parameters

Within the next sections, we aim to clarify the type of homogenization that it is aimed within the micromorphic framework and to draw distinctions with conventional static homogenization techniques. To this end, we analyze each elastic material parameter in detail, with the goal of providing a physically meaningful interpretation of the whole framework.

2.8.1 The general static picture

In the static case, the identification of material parameters for enriched continua is still today a highly non-trivial and largely unresolved problem with no universally accepted homogenization strategy accepted. Several methodologies have been previously proposed to upscale the heterogeneous model with microstructure to certain generalized model. In particular, homogenization towards Cosserat media has been addressed in (Hütter, 2019; Jeong et al., 2009), and homogenization procedures involving the classical Eringen–Mindlin micromorphic continuum have been explored in (Alavi et al., 2021; Biswas and Poh, 2017; Hütter, 2017, 2019; Forest, 2002; Rokoš et al., 2019, 2020a,b; Zhi et al., 2022). However, all these studies raise fundamental questions regarding the identification and interpretation of additional material parameters, most of them often within the framework of first-order computational homogenization (Trinh et al., 2012; Ganghoffer et al., 2023).

Within the relaxed micromorphic framework, a fine identification procedure has been previously proposed. In Neff et al. (2020), it is shown that the tensor C_{relax} (in their study called C_{micro}) can be identified as the Löwner half-order matrix supremum of suitable apparent stiffness tensors at the microscale. This means that C_{micro} can be understood as the stiffest admissible elastic response of the unit cell under affine Dirichlet boundary conditions. On the other hand, the macroscopic tensor C_{macro} follows from standard periodic homogenization and determines the mesoscopic elasticity tensor C_{rel} through the exact micro–macro consistency condition (eq. (20)). Both C_{micro} and C_{macro} are thus obtained via first-order numerical homogenization at the unit-cell level (Neff et al., 2020; Sarhil et al., 2023b; d’Agostino et al., 2022; Sarhil et al., 2024; Sky et al., 2024, 2025). This interpretation shows consistent limiting behavior. For a homogeneous unit cell, the identity $C_{\text{micro}} = C_{\text{macro}}$ implies $C_{\text{rel}} \rightarrow \infty$, and the micromorphic model reduces to classical linear elasticity. In contrast, the classical Eringen–Mindlin model degenerates into a second-gradient formulation with unbounded stiffness. We show here that this is false.

Within a different notation used for the relaxed micromorphic theory, note that the static tensorial consistency condition eq. (20), can be written as follows

$$\mathbb{C}_{\text{macro}} = \left(\frac{1}{\mathbb{C}_{\text{rel}}} + \frac{1}{\mathbb{C}_{\text{relax}}} \right)^{-1}, \quad (31)$$

where the macroscopic stiffness tensor $\mathbb{C}_{\text{macro}}$ comes from the interaction between $\mathbb{C}_{\text{rel}}$ and $\mathbb{C}_{\text{relax}}$, and formally resembles a harmonic-type combination at the scalar level (Neff et al., 2020; Sarhil et al., 2023a,b, 2024; Sky et al., 2024, 2025).

2.8.2 The general dynamical picture

In contrast to the previous presented static case, the dynamical regime introduces a frequency-dependent effective response due to micro-inertia effects. As a consequence, the macroscopic stiffness $\mathbb{C}_{\text{macro}}$ cannot be described by a single constant elasticity tensor, but shows different frequency-dependent asymptotic limits.

For both longitudinal and transverse polarizations, the macro-micro eq. (18) leads to the two limits

$$\lim_{\omega \rightarrow 0} \mathbb{C}_{\text{macro}} = \mathbb{C}_{\text{rel}} \mathbb{C}_{\text{relax}} (\mathbb{C}_{\text{rel}} + \mathbb{C}_{\text{relax}})^{-1}, \quad \lim_{\omega \rightarrow \infty} \mathbb{C}_{\text{macro}} = \mathbb{C}_{\text{rel}}. \quad (32)$$

The low-frequency limit ($\omega \rightarrow 0$) recovers the static effective response (eq. (20)). In contrast, in the high-frequency regime, the macroscopic response is only governed by the elasticity tensor $\mathbb{C}_{\text{rel}}$, that simply reflects the inability of the microstructure to fully relax. This transition between the two regimes highlights the role of internal length scales and micro-inertia in the micromorphic framework, which are absent in classical static homogenization approaches.

If we denote the lower limit $\lim_{\omega \rightarrow 0} \mathbb{C}_{\text{macro}} = \mathbb{C}_0$ and the upper limit $\lim_{\omega \rightarrow \infty} \mathbb{C}_{\text{macro}} = \mathbb{C}_{\text{rel}} = \mathbb{C}_\infty$, we can write the following expression

$$\mathbb{C}_{\text{relax}} = (\mathbb{C}_\infty - \mathbb{C}_0)^{-1} \mathbb{C}_0 \mathbb{C}_\infty - \eta \omega^2 \mathbb{1}. \quad (33)$$

It is then simply to observe that $\mathbb{C}_{\text{relax}}$ represents a hidden scale between $\mathbb{C}_0$ and $\mathbb{C}_\infty$ with a pole at $\mathbb{C}_0 = \mathbb{C}_\infty$. In other words, $\mathbb{C}_{\text{relax}}$ represents an intermediate scale governing the transition between the low-frequency and high-frequency responses, with a singular behavior as $\mathbb{C}_0 \rightarrow \mathbb{C}_\infty$.

Additionally, we can observe that eq. (33) encodes standard idea in rheology and viscoelasticity, i.e., the difference between $\mathbb{C}_0$ and $\mathbb{C}_\infty$ originates from internal (micro) relaxation mechanisms. Within this interpretation, $\mathbb{C}_{\text{relax}}$ measures the ability of the microstructure to relax and the term $\eta \omega^2$ acts like a frequency-dependent stiffening. Consequently, $\mathbb{C}_{\text{relax}}$ controls the separation between the relaxed and unrelaxed responses, i.e., how far $\mathbb{C}_0$ lies below $\mathbb{C}_\infty$.

This formal analogy with classical relaxation-type constitutive models, in which the material response transitions between relaxed and unrelaxed moduli, can be illustrated with the standard linear solid (Zener) model of viscoelasticity. The frequency-dependent viscoelastic modulus can be written as follows (Moczo et al., 2014)

$$\mathbb{C}_0 = \mathbb{C}_\infty \frac{1 + i \omega \tau_\varepsilon}{1 + i \omega \tau_\sigma}, \quad (34)$$

which describes a transition between a relaxed modulus $\mathbb{C}_0$ and an unrelaxed modulus $\mathbb{C}_\infty$, governed by the strain relaxation time τ_ε and stress relaxation time τ_σ (related to the dissipative mechanisms of the material under study).

It is crucial to note that eq. (33) shows a similar mathematical structure to eq. (34), with $\mathbb{C}_0$ and $\mathbb{C}_\infty$ representing the low- and high-frequency limits, respectively. However, the underlying mechanism is fundamentally different in both cases; while the viscoelastic relaxation is associated with dissipative mechanisms and the loss of energy, the micromorphic frequency dependent behavior originates from micro-inertia rather than dissipation, resulting in a purely real and non-dissipative response.

Laboratory experiments The micro-macro eq. (18), reveals the two limiting frequency cases given by eqs. (32). These limits give us information that can be obtained from laboratory experiments. At high frequencies we are able to directly obtain the elastic tensor $\mathbb{C}_{\text{rel}}$ and at low frequencies the combination $\mathbb{C}_{\text{rel}} \mathbb{C}_{\text{relax}} (\mathbb{C}_{\text{rel}} + \mathbb{C}_{\text{relax}})^{-1}$. From the low frequency expression, the value of $\mathbb{C}_{\text{relax}}$ can be readily determined after having obtained $\mathbb{C}_{\text{rel}}$. This gives us a clear physical picture given in Fig. 2, from which we can simply observe that the term $\mathbb{C}_{\text{relax}}$ can be understood as the internal stiffness associated with the microstructural relaxation responsible for the observed separation between the low- and high-frequency limits (eq. (33)).

The gap between high- and low-frequency moduli We define the frequency-dependent difference as

$$\Delta\mathbb{C}(\omega) := \mathbb{C}_{\infty} - \mathbb{C}_{\text{macro}}(\omega). \quad (35)$$

Since $\mathbb{C}_{\infty} = \mathbb{C}_{\text{rel}}$, substitution of eq. (18) and using $\eta = \rho L_c^2$ gives

$$\Delta\mathbb{C}(\omega) = \mathbb{C}_{\text{rel}} \left(\mathbb{C}_{\text{rel}} + \mathbb{C}_{\text{relax}} - \rho L_c^2 \omega^2 \mathbb{1} \right)^{-1} \mathbb{C}_{\text{rel}}. \quad (36)$$

This frequency-dependent difference possesses a pole whenever

$$\mathbb{C}_{\text{rel}} + \mathbb{C}_{\text{relax}} - \rho L_c^2 \omega^2 \mathbb{1} = 0. \quad (37)$$

and it does not, in general, represent a smooth bounded transition between the low- and high-frequency elastic moduli.

The fixed gap between the two asymptotic limits is instead defined by

$$\Delta\mathbb{C}_{\text{asym}} := \mathbb{C}_{\infty} - \mathbb{C}_0 = \mathbb{C}_{\text{rel}} (\mathbb{C}_{\text{rel}} + \mathbb{C}_{\text{relax}})^{-1} \mathbb{C}_{\text{rel}}. \quad (38)$$

Hence, $\mathbb{C}_{\text{relax}}$ controls the separation between the relaxed and unrelaxed asymptotic responses. In particular, From which we can observe that

- **small** $\mathbb{C}_{\text{relax}} \rightarrow 0$: $\Delta\mathbb{C}_{\text{asym}} \rightarrow \infty \Rightarrow$ large gap, strong relaxation.
- **large** $\mathbb{C}_{\text{relax}} \rightarrow \infty$: $\Delta\mathbb{C}_{\text{asym}} \rightarrow 0 \Rightarrow$ small gap, weak relaxation.

This shows that the asymptotic stiffness gap between the relaxed ($\mathbb{C}_0$) and unrelaxed ($\mathbb{C}_{\infty}$) elastic responses decreases as $\mathbb{C}_{\text{relax}}$ increases. Therefore, $\mathbb{C}_{\text{relax}}$ can be interpreted as the material parameter controlling the separation between the low- and high-frequency elastic moduli. Small values of $\mathbb{C}_{\text{relax}}$ correspond to a pronounced difference between the two asymptotic regimes (strong microstructural relaxation), whereas large values of $\mathbb{C}_{\text{relax}}$ reduce this difference until both limits eventually coincide. This interpretation is schematically illustrated in Fig. 2.

Interpretation of the relaxation parameter $\mathbb{C}_{\text{relax}}$ Although the physical meaning of $\mathbb{C}_{\text{relax}}$ is clear, it can be obscure/unclear to understand what a large or small $\mathbb{C}_{\text{relax}}$ will produce large or small gaps in practice. In simpler words, how do we know that we are in presence of a large or a small $\mathbb{C}_{\text{relax}}$, that can be physically translated into large or small relaxation? To answer this question and to gain complete physical insight, we next define, based on eq. (36), the dimensionless gap between high- and low-frequency moduli as follows

$$\langle \Delta\mathbb{C}, \mathbb{C}_{\infty}^{-1} \rangle = \left\langle \left(\mathbb{C}_{\text{rel}} + \mathbb{C}_{\text{relax}} - \rho L_c^2 \omega^2 \mathbb{1} \right)^{-1}, \mathbb{C}_{\text{rel}} \right\rangle. \quad (39)$$

This dimensionless gap $\langle \Delta\mathbb{C}, \mathbb{C}_{\infty}^{-1} \rangle$ is a practical measure of the magnitude of relaxation at a given frequency relative to the instantaneous stiffness. By inspection of the normalized gap in eq. (39), one can define representative illustrative regimes of relaxation that provide a practical interpretation of the model response. While no universal thresholds exist to classify the strength of relaxation, the normalized gap in Eq. (39) allows representative regimes to be defined for a practical interpretation of the material response.

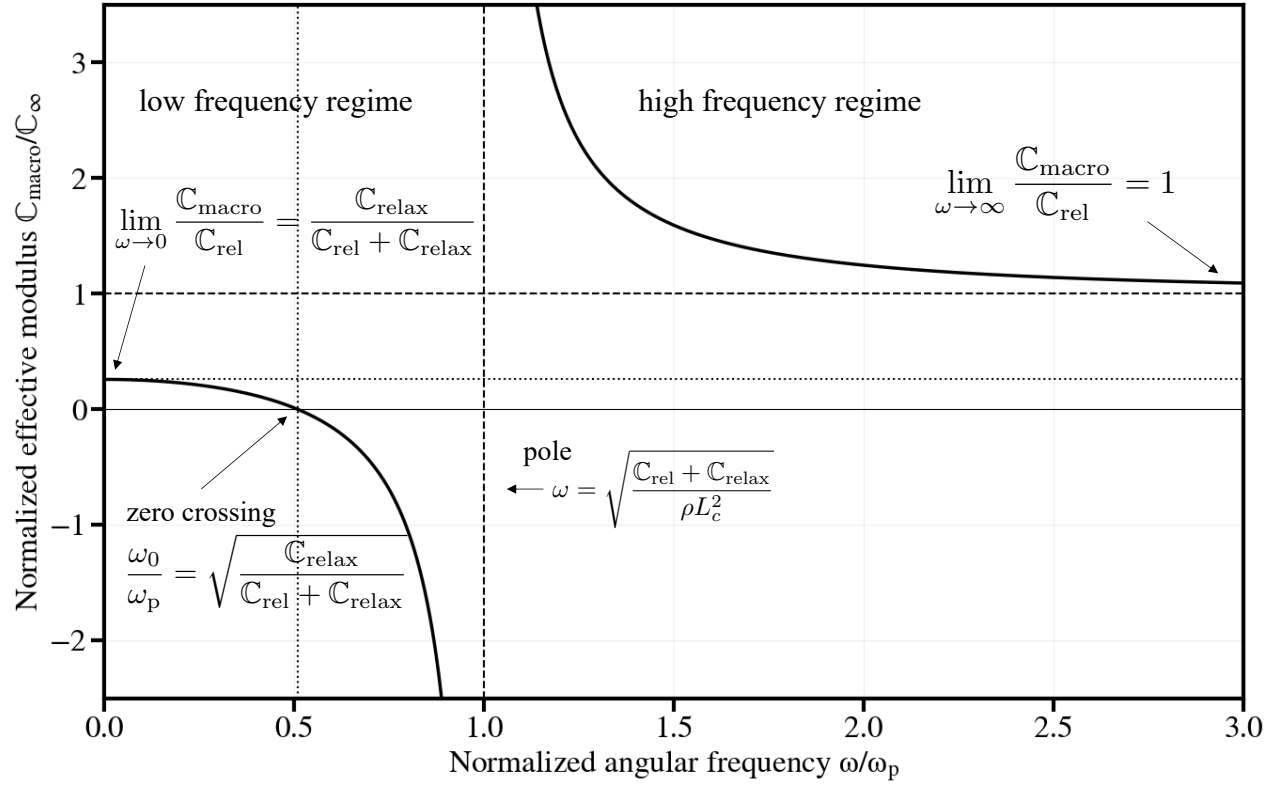


Figure 2: Normalized effective elastic modulus given by eq. (18) as a function of the normalized angular frequency. The horizontal asymptotes correspond to the low- and high-frequency limits, while the vertical asymptote identifies the pole.

2.8.3 The general dynamical picture: low frequency regime

We next analyze the low frequency regime ($\omega \rightarrow 0$), by neglecting the characteristic length $\mathbb{L}_c$ and $\mathbb{C}_{\text{rot}} = \mathbb{C}_{\text{spin}} = 0$ in the micromorphic energy (eq. (4)), which reduces to

$$\mathcal{E}_{\text{total}} = \frac{1}{2} \underbrace{\langle \text{sym}(\nabla u - P), \mathbb{C}_{\text{rel}} \text{sym}(\nabla u - P) \rangle}_{\text{relative deformation energy}} + \frac{1}{2} \underbrace{\langle \text{sym} P, \mathbb{C}_{\text{relax}} \text{sym} P \rangle}_{\text{relaxation resistance energy}}. \quad (40)$$

The tensor $\mathbb{C}_{\text{rel}}$ governs the elastic energy associated with the mismatch between the macroscopic deformation ∇u and the micro-distortion P , through the relative deformation measure $\text{sym}(\nabla u - P)$. In contrast, the tensor $\mathbb{C}_{\text{relax}}$ governs the energetic resistance to internal deformation modes by penalizing the activation of the micro-distortion field through the term $\text{sym} P$.

Relationship between ∇u and P The minimization of the functional in eq. (40) with respect to the field variables (u, P) , gives the following relation between ∇u and P

$$P = \mathbb{C}_{\text{rel}} (\mathbb{C}_{\text{rel}} + \mathbb{C}_{\text{relax}})^{-1} \nabla u, \quad (41)$$

which means that, in the static case, $P \leq \nabla u$, i.e., the magnitude of P is always proportional to the magnitude of ∇u and never larger. From eq. (41), we can find the next four limits

$$\lim_{\mathbb{C}_{\text{relax}} \rightarrow 0} P = \nabla u, \quad \lim_{\mathbb{C}_{\text{relax}} \rightarrow \infty} P = 0, \quad \lim_{\mathbb{C}_{\text{rel}} \rightarrow 0} P = 0, \quad \lim_{\mathbb{C}_{\text{rel}} \rightarrow \infty} P = \nabla u. \quad (42)$$

For fixed material parameters $(\mathbb{C}_{\text{rel}}, \mathbb{C}_{\text{relax}})$, three distinct regimes can be identified:

1. **fully coupled regime:** $P \approx \nabla u$. The micro-distortion follows the macroscopic deformation, producing a negligible relative deformation, i.e., $\nabla u - P \approx 0$. This corresponds to the case where internal relaxation is almost absent $\mathbb{C}_{\text{relax}} \approx 0$.
2. **suppressed microstructure:** $P \approx 0$. The microstructural deformation is inhibited, and the macroscopic deformation is almost entirely accommodated through the relative deformation. This corresponds to the case where internal deformation is strongly penalized by a large relaxation stiffness $\mathbb{C}_{\text{relax}}$.
3. **intermediate regime:** $P < \nabla u$. The internal deformation P partially follows the macroscopic deformation ∇u , which leads to a finite mismatch between the two fields.

These limits demonstrate that $\mathbb{C}_{\text{rel}}$ and $\mathbb{C}_{\text{relax}}$ jointly determine how deformation is sheared between the macroscopic field ∇u and the internal micro-distortion field P . A small $\mathbb{C}_{\text{relax}}$ allows the microstructure to closely follow the macroscopic deformation, whereas a large $\mathbb{C}_{\text{relax}}$ suppresses the development of internal deformation. Similarly, $\mathbb{C}_{\text{rel}}$ controls the energetic cost associated with the mismatch between the two fields, thereby enforcing their kinematic compatibility. In contrast to the static case, the dynamic case emerges from the competition between the mismatch resistance governed by $\mathbb{C}_{\text{rel}}$ and the relaxation resistance governed by $\mathbb{C}_{\text{relax}}$. This balance is further modified by micro-inertia, which determines whether the local deformation P can respond to changes in the macroscopic deformation ∇u over a different frequency regime.

2.8.4 The general dynamical picture: high frequency regime

Including the kinetic energy of the local deformation in eq. (40), and setting $L_c = \mathbb{C}_{\text{rot}} = \mathbb{C}_{\text{spin}} = 0$, leads to the following expression

$$\mathcal{E}_{\text{total}} = \frac{1}{2} \underbrace{\langle \text{sym}(\nabla u - P), \mathbb{C}_{\text{rel}} \text{sym}(\nabla u - P) \rangle}_{\text{relative deformation energy}} + \frac{1}{2} \underbrace{\langle \text{sym} P, \mathbb{C}_{\text{relax}} \text{sym} P \rangle}_{\text{relaxation resistance energy}} + \underbrace{\frac{1}{2} \langle \partial_t P, \eta \partial_t P \rangle}_{\text{microstructure kinetic energy}}. \quad (43)$$

The minimization of the functional in eq. (43) with respect to the field variables (u, P) , and assuming plane wave solutions, gives the following frequency-dependent relation between ∇u and P

$$\hat{P} = \mathbb{C}_{\text{rel}} \left(\mathbb{C}_{\text{rel}} + \mathbb{C}_{\text{relax}} - \eta \omega^2 \mathbb{1} \right)^{-1} \nabla \hat{u}, \quad (44)$$

where $\hat{P}$ and $\hat{u}$ denote the amplitudes of the corresponding plane waves for P, u respectively. In the high-frequency limit $\eta \omega^2 \mathbb{1} \gg \mathbb{C}_{\text{rel}} + \mathbb{C}_{\text{relax}}$, we obtain

$$\hat{P} \approx \nabla \hat{u} \rightarrow 0, \quad (45)$$

which means that the local deformation P is strongly suppressed by inertia. Within this regime, the microstructure cannot follow the macroscopic deformation and its contribution thus becomes negligible. This behavior indicates that, at high frequencies, deformation is carried almost entirely by the macroscopic field u , leading to an effective response governed by the stiffness $\mathbb{C}_{\text{rel}}$.

2.8.5 The characteristic length: $L_c \rightarrow \eta \partial_t^2 P$

The characteristic length L_c defines an intrinsic length scale associated with the inertial response of the microstructure through the relation $\eta = \rho L_c^2$. Therefore, L_c controls the dynamic resistance to acceleration and as a consequence, it does not represent a directly measurable geometric variable such as the grain size, but rather as a constitutive length scale governing the inertial response of the internal deformation modes.

Its influence in wave propagation emerges through the combined scaling with frequency $L_c^2 \omega^2$, which represents a competition between microstructural inertia and elastic stiffness. In particular, one can observe dispersion effects when the wavelength becomes comparable to the effective length scale associated with L_c , or equivalently when $L_c^2 \omega^2$ becomes comparable to the total stiffness $(\mathbb{C}_{\text{rel}} + \mathbb{C}_{\text{relax}})$, see eq. (18). When this happens, the propagating wave interacts dynamically with the microstructure, leading to frequency-dependent effective properties.

Thus, L_c should be interpreted as the constitutive length scale that determines the frequencies (or wavelengths) at which the inertial response of the microstructure becomes significant.

2.8.6 The takeaway

The micromorphic model can be interpreted as a form of frequency-dependent homogenization, in which the effective macroscopic stiffness emerges from the dynamic interaction between the macroscopic deformation (∇u) and internal/local deformation (P). The model does not attempt to identify the microstructure itself, but rather its mechanical effect on the macroscopic response. Unlike classical static homogenization, where effective properties depend solely on the geometry of the microstructure, the frequency dependence of the micromorphic model arises from the inertia and dynamics of the microstructural/local deformation field P . As a result, the material exhibits a dispersive effective response without requiring geometric constraints. In this sense, the model provides a dynamical average of the microstructural dynamics, where the effective properties depend on frequency due to the presence of internal inertia.

The elastic tensors $\mathbb{C}_{\text{rel}}$ and $\mathbb{C}_{\text{relax}}$ do not represent literal phase stiffnesses. Instead, they quantify the energetic resistance of macroscopic deformation versus microstructural/local deformation. The effective macroscopic stiffness $\mathbb{C}_{\text{macro}}$ emerges from the dynamic coupling between ∇u and P , leading to frequency-dependent behavior and wave dispersion (see Fig. 3).

An important consequence is that different microstructural configurations such as fractured media, porous media with fluid- or gas-saturated regions, soft/hard inclusions, etc., many may lead to the same effective response $\mathbb{C}_{\text{macro}}$ (see Fig. 3). Within this framework, $\mathbb{C}_{\text{relax}}$ should be interpreted as a relaxation parameter: it quantifies the resistance of the internal microstructural deformation and the amount of stress that can be relaxed or accommodated by the microstructure. Microstructural features that are compliant, soft, or capable of rearrangement all contribute to a lower $\mathbb{C}_{\text{relax}}$, while stiffer or more constrained internal structures produce a higher $\mathbb{C}_{\text{relax}}$. This means that $\mathbb{C}_{\text{relax}}$ does not directly identify the physical composition neither geometry of the microstructure but it provides an effective measure of the collective relaxation capacity of the microstructure.

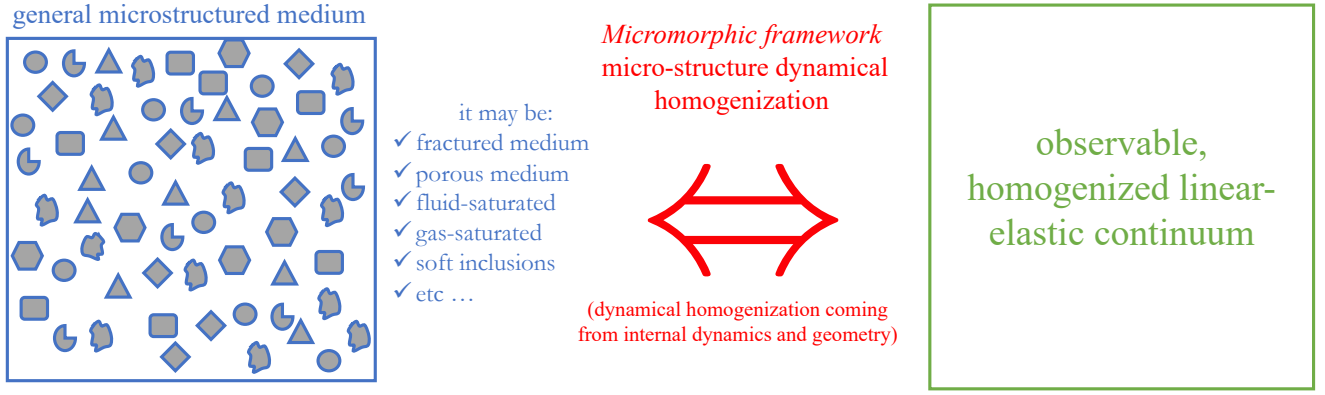


Figure 3: Different microstructural configurations (e.g., cracks, pores with fluid/gas filled regions, or hard/soft inclusions) can lead to indistinguishable macroscopic responses within the micromorphic framework. The model captures the effective stiffness of the internal deformation mechanism rather than the specific physical nature of the microstructure.

3 Non-linear shear softening

Several soils and/or rocks with different type of microstructure may undergo shear softening in presence of moderate to strong shear deformation. Shear softening simply refers to the reduce of the effective shear stiffness of the material. Capturing this behavior is of primordial importance for predicting phenomena such as liquefaction, strain localization, or damping in geomaterials (Hardin, 1965; Lemaitre, 1992; Ishihara, 1996). However, despite its importance and important consequences in seismological applications (e.g. Hartzell et al., 2004; Esfahani et al., 2024; Ji et al., 2021; Guéguen et al., 2019; Gélis and Bonilla, 2012; Lai et al., 2025), the theoretical modeling of these kind of effects are commonly done relying on purely empirical hyperbolic or polynomial softening laws (Hardin and Drnevich, 1972; Pyke, 1979; Hartzell et al., 2004; Beresnev and Wen, 1996; Ishihara, 1996; Darendeli, 2001).

Our goal in this section is to capture fundamental physics of how microstructural mechanisms such as grain rearrangements, contact degradation, and micro-crack activation progressively reduce the material shear stiffness under increasing shear strain. As we will see, unlike purely empirical hyperbolic laws, we aim to link macroscopic softening directly to fundamentally physical meaningful microstructural mechanics, allowing for a possible more realistic (or less empirical) representation of material behavior under large shear deformation. To do so, we next assume that the average shear properties of the material suffers any type of mechanical degradation as follows (Lemaitre, 1992)

$$\mu(\gamma) = \mu^0 (1 - d(\gamma)), \quad (46)$$

where μ^0 is the initial average shear modulus and we introduce a scalar variable $d(\gamma)$ as a function of the average shear strain γ that represents the average shear degradation of the material as follows (Lemaitre, 1992)

$$0 \leq d(\gamma) \leq 1, \quad (47)$$

where $d = 0$ is the intact material and $d = 1$ is fully degraded material. In physical terms d may represent crack density, contact degradation, grain debonding, and/or micro-slip activation, etc (Lemaitre, 1992).

A physically reasonable evolution law is an exponential activation, consistent with classical continuum damage mechanics formulations (Lemaitre, 1992), as follows

$$d(\gamma) = 1 - \exp\left(-\frac{\gamma}{\gamma_c}\right), \quad (48)$$

where γ_c is a characteristic shear strain controlling, for example, crack activation. Substituting $d(\gamma)$ into eq. (46) we can write a strain dependent shear elastic parameter as follows

$$\mu(\gamma) = \mu^0 \exp\left(-\frac{\gamma}{\gamma_c}\right). \quad (49)$$

We next interpret the shear modulus $\mu(\gamma)$ defined in eq. (46) as the effective modulus measured at the scale of laboratory experiments. Such measurements are typically performed at relatively high frequencies, for which internal microstructural mechanisms do not have sufficient time to relax. As a result, in our interpretation, $\mu(\gamma)$ represents a high-frequency (unrelaxed) response of the material, in contrast to the lower-frequency behavior observed at seismic/seismological scales, where internal relaxation processes may significantly reduce much more the effective stiffness.

Within the micromorphic framework, this high-frequency response is identified with the elastic coupling modulus μ_{rel} . Therefore, we set $\mu_{\text{rel}}(\gamma) = \mu(\gamma)$, and the micromorphic consistency condition (eq. (??)) can be written as follows

$$\mu_{\text{macro}}(\gamma) = \frac{\mu_{\text{relax}} \left[\mu_{\text{rel}}^0 \exp(-\gamma/\gamma_c) - \eta \omega^2 \right]}{\mu_{\text{relax}} + \mu_{\text{rel}}^0 \exp(-\gamma/\gamma_c) - \eta \omega^2}, \quad (50)$$

which produces nonlinear softening behavior. Note that for small strain and low frequencies ($\gamma, \omega \rightarrow 0$), we obtain

$$\lim_{\gamma, \omega \rightarrow 0} \mu_{\text{macro}} = \frac{\mu_{\text{rel}}^0 \mu_{\text{relax}}}{\mu_{\text{rel}}^0 + \mu_{\text{relax}}}. \quad (51)$$

Thus, the small-strain stiffness is finite and physically meaningful. For shear large strains ($\gamma \rightarrow \infty$) and low frequencies ($\omega \rightarrow 0$) we have

$$\lim_{\gamma \rightarrow \infty, \omega \rightarrow 0} \mu_{\text{macro}} = 0, \quad (52)$$

thus the material losses shear stiffness due to, for example, progressive crack growth. In physical terms, we can say that shear strain activates micro-cracks and crack growth reduces microstructure stiffness. The resulting reduced micro stiffness lowers the effective macroscopic modulus and as a consequence, softening emerges from internal structural degradation rather than from a fitted empirical formula, as we will next see.

Note that the strain-dependent internal resonance frequency in eq. (50) is given by

$$\omega_p(\gamma) = \sqrt{\frac{\mu_{\text{relax}} + \mu_{\text{rel}}^0 \exp\left(-\frac{\gamma}{\gamma_c}\right)}{\eta}}. \quad (53)$$

This frequency corresponds to the natural oscillation of the internal microstructural degree of freedom. Since the coupling modulus $\mu_{\text{rel}}(\gamma)$ decreases with increasing shear strain, the resonance frequency also progressively shifts toward lower frequencies. As a consequence, progressive microstructural degradation not only reduces the effective shear stiffness but also lowers the characteristic frequency of the internal resonance. In the absence of dissipation, this resonance appears as a pole of the dynamically condensed effective modulus.

3.1 Relationship with the Hardin–Drnevich modulus reduction law

The shear modulus decay with shear strain measured in the laboratory, is often represented with the following phenomenological Hardin–Drnevich modulus reduction curve equation (Hardin and Drnevich, 1972; Pyke, 1979; Hartzell et al., 2004; Beresnev and Wen, 1996; Ishihara, 1996)

$$\frac{\mu}{\mu_0} = \frac{1}{1 + \gamma/\gamma_{\text{ref}}} \quad (54)$$

where μ_0 is the undisturbed shear modulus, γ = macroscopic shear strain and $\gamma_{\text{ref}} = \tau_0/\mu_0$ is the reference strain controlling the onset of softening and τ_0 is the maximum shear stress that the material can support in the initial state.

To improve the fit to experimental data, several modifications to eq. (54) have been proposed such as, Darendeli's single parameter (α) model and a two-parameter or modified Darendeli's model (Darendeli, 2001; Zhang et al., 2005) given by the following expressions

$$\frac{\mu}{\mu_0} = \frac{1}{1 + \left(\frac{\gamma}{\gamma_{\text{ref}}}\right)^\alpha}, \quad \frac{\mu}{\mu_0} = \frac{1}{1 + \beta \left(\frac{\gamma}{\gamma_{\text{ref}}}\right)^\alpha}, \quad (55)$$

where α, β are phenomenological fitting parameters.

The set of phenomenological eqs. (54)–(55) are simple and, in general, are able to match experimental observations. However, they are completely phenomenological and no physical interpretation of the microstructure is given. As an attempt to solve these limitations, we next introduce a micromorphic formulation framework based in eq. (50). To do so, we consider the regime of moderate strains and expand the exponential term in a first-order Taylor approximation in eq. (49) as follows

$$\mu_{\text{rel}}(\gamma) = \mu_{\text{rel}}^0 \exp\left(-\frac{\gamma}{\gamma_c}\right) \approx \mu_{\text{rel}}^0 \frac{1}{1 + \gamma/\gamma_c}. \quad (56)$$

Substituting this approximation into the consistency condition eq. (50), leads to the following expression

$$\frac{\mu_{\text{macro}}}{\mu_{\text{rel}}^0} = \frac{\mu_{\text{relax}}}{\mu_{\text{rel}}^0} \frac{\mu_{\text{rel}}^0 - \eta\omega^2 \left(1 + \frac{\gamma}{\gamma_c}\right)}{\mu_{\text{rel}}^0 + (\mu_{\text{relax}} - \eta\omega^2) \left(1 + \frac{\gamma}{\gamma_c}\right)}. \quad (57)$$

Note that eq. (57) is simply the ratio between the low frequency limit μ_{macro} and high frequency limit μ_e (see Section 2.8.2). Eq. (57) has the next two frequency limits

$$\lim_{\omega \rightarrow 0} \frac{\mu_{\text{macro}}}{\mu_{\text{rel}}^0} = \frac{\mu_{\text{relax}}}{\mu_{\text{rel}}^0 + \mu_{\text{relax}}} \frac{1}{1 + \frac{\mu_{\text{relax}}}{\mu_{\text{rel}}^0 + \mu_{\text{relax}}} \gamma/\gamma_c}, \quad \lim_{\omega \rightarrow \infty} \frac{\mu_{\text{macro}}}{\mu_{\text{rel}}^0} = \frac{\mu_{\text{relax}}}{\mu_{\text{rel}}^0}. \quad (58)$$

At low frequencies, the term $\mu_{\text{relax}}/(\mu_{\text{rel}}^0 + \mu_{\text{relax}})$ sets the *maximum relative macroscopic modulus*, i.e., the value of $\mu_{\text{macro}}/\mu_{\text{rel}}^0$ at zero strain ($\gamma = 0$). Taking the limit of vanishing shear strain, $\gamma \rightarrow 0$, in eq. (57) gives

$$\lim_{\gamma \rightarrow 0} \frac{\mu_{\text{macro}}}{\mu_e^0} = \frac{\mu_{\text{relax}}}{\mu_e^0 + \mu_{\text{relax}} - \eta\omega^2} \left(1 - \frac{\eta\omega^2}{\mu_e^0}\right), \quad (59)$$

which is finite. In this limit, $\mu_e \rightarrow \mu_e^0$ (see eq. (49)), and the energetic penalty associated with the relative deformation ($\nabla u - P$) reaches its maximum value. As a consequence, the micromorphic model tends to minimize the relative deformation between the macroscopic and microstructural kinematics. The extent to which this occurs, however, is controlled by the competition between the coupling stiffness μ_e^0 and the relaxation stiffness μ_{relax} .

It is important to emphasize that μ_{relax} does not represent the stiffness of a particular microstructural phase. Instead, it quantifies the energetic resistance for activating the internal microdeformation field. Large values of μ_{relax} suppress the development of P , whereas smaller values allow the microstructure to accommodate a larger fraction of the imposed deformation.

Within this interpretation, the Hardin–Drnevich empirical law (eq. (54)) may be regarded as the macroscopic manifestation of a progressive activation of internal relaxation mechanisms governed by the strain-dependent coupling parameter $\mu_e(\gamma)$. Within the micromorphic framework, relaxation is not governed by a softening of a material phase, but by the progressive release of kinematic constraints between coupled deformation mechanisms.

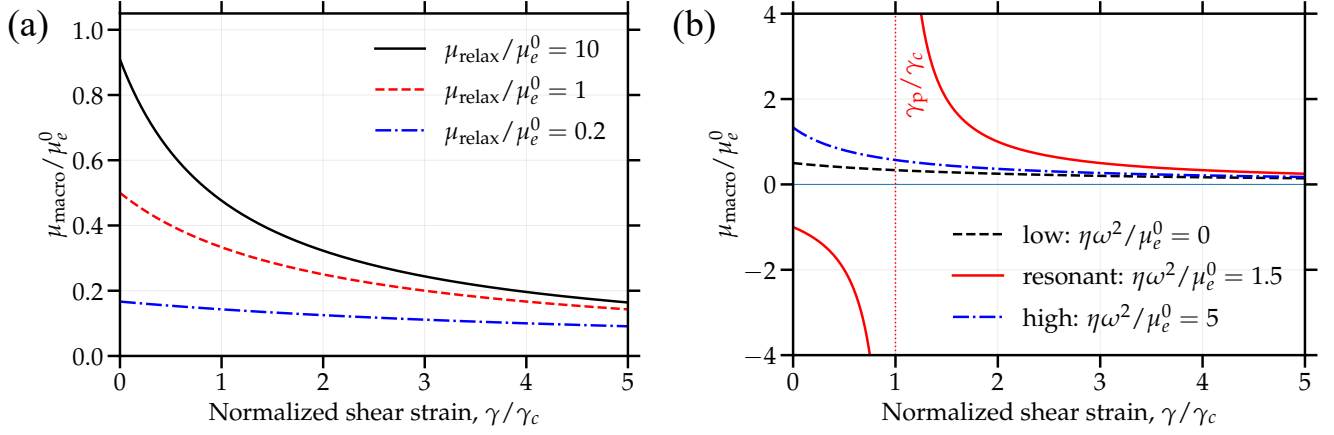


Figure 4: (a) Comparison of macroscopic softening curves. (b) Effect of frequency on the macroscopic softening response.

In Fig. 4–a, when $\mu_{\text{relax}} \gg \mu_{\text{rel}}^0$ (blue curve), the material starts close to its unrelaxed stiffness. In this regime, the large relaxation stiffness strongly penalizes the activation of the internal microdeformation field, so that $P \ll \nabla u$. As a consequence, only a limited amount of the applied deformation is accommodated by the microstructure, and the macroscopic response remains comparatively stiff.

When $\mu_{\text{rel}}^0 \sim \mu_{\text{relax}}$ (red dashed curve in Fig. 4–a), the internal microdeformation allows a larger fraction of the imposed deformation to be relaxed internally. The effective macroscopic modulus is therefore reduced, and the strain-dependent softening becomes more pronounced.

Figure 4–b illustrates the influence of frequency. The micro-inertial contribution modifies the dynamic coupling between the macroscopic deformation and the internal microstructure. As the excitation frequency increases, the effective modulus evolves toward the characteristic internal resonance of the micromorphic medium. In the case where no viscoelastic effects are present, the effective modulus first decreases, vanishes at the characteristic frequency $\omega_z = \sqrt{\mu_{\text{relax}}/\eta}$, and subsequently exhibits a pole at $\omega_p = \sqrt{(\mu_{\text{rel}}^0 + \mu_{\text{relax}})/\eta}$. Beyond this resonant regime, the microdeformation field is no longer able to follow the macroscopic motion, and the material response approaches the unrelaxed coupling stiffness $\mu_{\text{rel}}(\gamma)$.

The resonant behavior predicted by the present formulation is a direct consequence of the dynamics of P , i.e., $\partial_t^2 P$. However, in real scenarios, dissipative mechanisms are expected to regularize this singular behavior, producing a smooth transition between low and high frequency regimes..

Equation (57) shows the interplay between the strain-dependent coupling stiffness, the resistance to internal relaxation, and the frequency-dependent inertial effects associated with the internal degrees of freedom. The general physical picture can be summarized as follows:

- The term

$$\frac{\mu_{\text{relax}} - \eta\omega^2}{\mu_{\text{rel}}^0 + \mu_{\text{relax}} - \eta\omega^2} \quad (60)$$

determines the normalized effective stiffness in the limit of vanishing shear strain. At low frequencies, it reflects the competition between the coupling stiffness μ_{rel}^0 and the resistance to internal deformation μ_{relax} . The micro-inertial contribution modifies this balance as the frequency approaches the characteristic internal frequencies of the material.

- The hyperbolic-like term

$$\frac{1}{1 + \frac{\mu_{\text{relax}} - \eta\omega^2}{\mu_{\text{rel}}^0 + \mu_{\text{relax}} - \eta\omega^2} \frac{\gamma}{\gamma_c}} \quad (61)$$

governs the reduction of the effective modulus with increasing shear strain. This behavior originates from the strain-dependent degradation of the coupling stiffness $\mu_{\text{rel}}(\gamma)$. The frequency modifies the apparent softening rate through the internal micro-inertia.

- In the low-frequency limit, $\omega \rightarrow 0$, the normalized effective modulus becomes

$$\lim_{\omega \rightarrow 0} \frac{\mu_{\text{macro}}}{\mu_{\text{rel}}^0} = \frac{\mu_{\text{relax}}}{\mu_{\text{rel}}^0 + \mu_{\text{relax}}} \frac{1}{1 + \frac{\mu_{\text{relax}}}{\mu_{\text{rel}}^0 + \mu_{\text{relax}}} \frac{\gamma}{\gamma_c}}. \quad (62)$$

In this regime, the internal degree of freedom can follow the macroscopic deformation and partially relax the imposed shear strain. The effective response is therefore softer than the unrelaxed coupling response.

- In the high-frequency limit, $\omega \rightarrow \infty$, the microdeformation amplitude tends to zero and

$$\lim_{\omega \rightarrow \infty} \mu_{\text{macro}} = \mu_{\text{rel}}(\gamma). \quad (63)$$

The internal deformation is dynamically constrained, and the material approaches the unrelaxed coupling stiffness. Unlike the previous interpretation, however, the strain dependence does not disappear in this limit because the coupling modulus itself has been assumed to degrade according to $\mu_{\text{rel}}(\gamma)$.

- In general terms, increasing frequency suppresses the internal relaxation associated with the microdeformation field and shifts the response from the relaxed modulus toward the unrelaxed coupling modulus.

Equation (57) therefore provides a description of the frequency- and strain-dependent material response. The model predicts a transition from a relaxation-dominated regime at low frequencies to an unrelaxed elastic regime at high frequencies. In the ideal lossless case, this transition is characterized by an internal resonance associated with the microstructural degree of freedom.

3.2 Experimental verification

We next match experimental observations to the introduced softening law (eq. (57)) using laboratory data from [Dammala et al. \(2019\)](#). The soil studied is a sand collected from the shore of Brahmaputra River near Guwahati city in Assam, India (a highly active seismic region). Based on the grain size distribution, the collected sand falls in the highly liquefiable materials category, classified as poorly graded sand according to unified soil classification system (USCS). The studied dynamic soil properties of these samples are useful for performing seismic design of structures or seismic re-qualification of existing structures in similar soils ([Dammala et al., 2019](#)).

Two types of apparatus (resonant column-RC and cyclic triaxial-CTX) were used to test the strain dependent dynamic properties and liquefaction potential of the samples. For our analysis, we use the variation of μ/μ_0 of the samples at 50% relative density and 50 kPa confining pressure. The reference strain value was considered from the experimental results at μ/μ_0 of 0.5.

To match the experimental data with the micromorphic model, we write eq. (57) as follows

$$\frac{\mu_{\text{macro}}}{\mu_{\text{rel}}^0} = A \frac{1 - C \left(1 + \frac{\gamma}{\gamma_c}\right)}{1 + B \frac{\gamma}{\gamma_c}}, \quad (64)$$

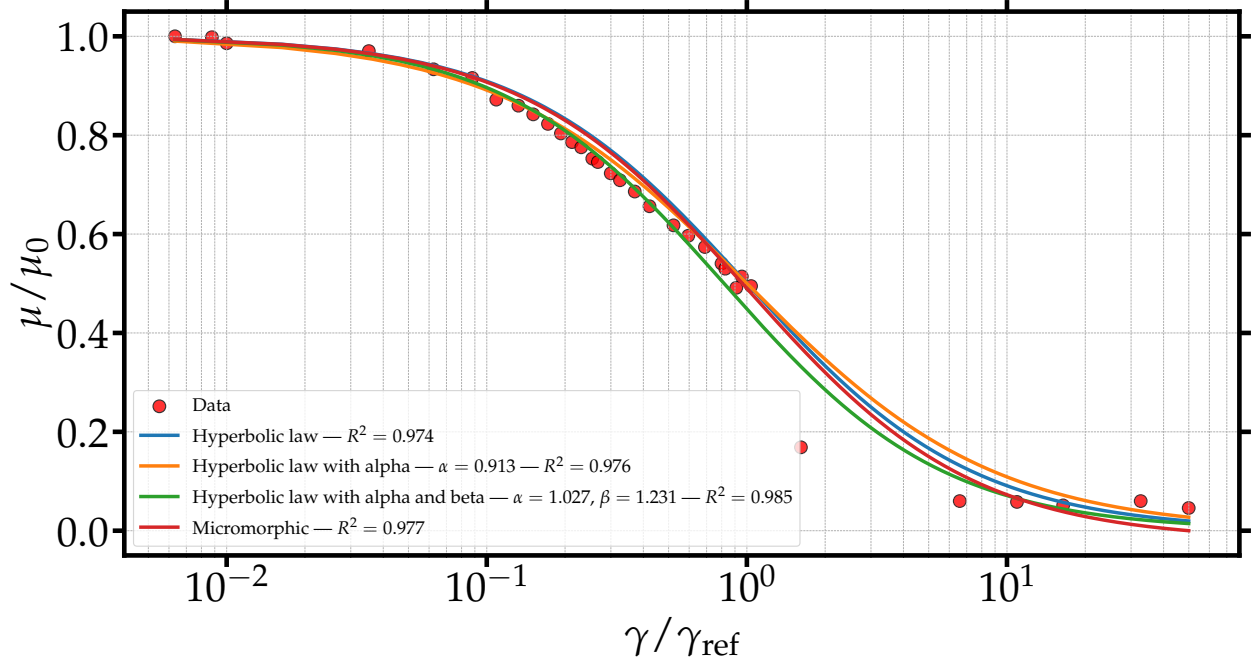


Figure 5: Soil data from the Brahmaputra River at 50% relative density and 50 kPa confining pressure (Dammala et al., 2019), fitted against the micromorphic model (eq. (57)) and all phenomenological hyperbolic laws (eqs. (54)–(55)).

where the effective parameters are defined as

$$A = \frac{\mu_{\text{relax}}}{\mu_{\text{relax}} + \mu_{\text{rel}}^0 - \eta \omega^2}, \quad B = \frac{\mu_{\text{relax}} - \eta \omega^2}{\mu_{\text{relax}} + \mu_{\text{rel}}^0 - \eta \omega^2}, \quad C = \frac{\eta \omega^2}{\mu_{\text{rel}}^0}. \quad (65)$$

The results of fitting the micromorphic (eq. (57)) and all hyperbolic (eqs. (54)–(55)) models to the experimental data are shown in Fig. 5. The best micromorphic fitted parameters are

$$A \simeq 1, \quad B \simeq 1, \quad C = 0.0196. \quad (66)$$

Note that the values of A and B have been rounded for presentation. However, their not-rounded values satisfy the consistency relation $A = B + C(1 - B)$.

Unlike traditional hyperbolic phenomenological laws, which provide purely empirical fits to experimental data (see Fig. 5), the micromorphic softening law (eq. (57)) allows for a physically grounded interpretation of the fitted parameters (A, B, C):

- $A \simeq 1$ indicates that the resistance to internal relaxation is large compared to both the coupling stiffness μ_{rel}^0 and the inertial contribution $\eta \omega^2$ ($\mu_{\text{relax}} \gg \mu_{\text{rel}}^0 - \eta \omega^2$). This means that, in the present sub-resonant regime where $\eta \omega^2 < \mu_{\text{relax}}$, internal deformation modes are weakly activated, and the material behaves close to its maximum stiffness at small strains.
- The fitted value $C = 0.0196$ shows that $\eta \omega^2 / \mu_{\text{rel}}^0 = 0.0196$. This means that the inertial contribution $\eta \omega^2$ is small under the tested conditions. This corresponds to a quasi-static (weakly dynamic) regime in which dynamic effects do not significantly influence the activation of deformation induced by microstructure, and far from the characteristic zero and pole predicted by the ideal lossless model.

- $B \approx 1$ indicates that $\mu_{\text{relax}} - \eta\omega^2$ is large compared with μ_{rel}^0 . Additionally, the difference between A and B is

$$A - B = \frac{\eta\omega^2}{\mu_{\text{relax}} + \mu_{\text{rel}}^0 - \eta\omega^2}. \quad (67)$$

Because the micro-inertial contribution is small in the present experiment, A and B take very similar values. The parameter B describes the combined influence of the resistance to internal deformation and micro-inertia on the strain-dependent denominator of the effective modulus.

The fitted parameters therefore indicate that the experimental response corresponds to a sub-resonant regime characterized by a strong resistance to internal microdeformation and a comparatively small micro-inertial contribution. The observed modulus reduction is mainly governed by the strain-dependent degradation of the coupling stiffness $\mu_{\text{rel}}(\gamma)$, rather than by proximity to an internal resonance.

4 Discussion

4.1 Softening laws

The micromorphic hyperbolic law eq. (57) shows that nonlinear softening originates from the progressive reduction of the coupling shear stiffness $\mu_{\text{rel}}(\gamma)$, which governs the interaction between macroscopic deformation and internal degrees of freedom. In this sense, $\mu_{\text{rel}}(\gamma)$ determines how strongly the macroscopic deformation constrains the local deformation. The elastic tensor μ_{relax} represents a relaxation parameter controlling the resistance to internal deformation modes. Material softening therefore arises from a progressive weakening of this constraint, allowing internal deformation mechanisms to become increasingly active. In addition, the characteristic length L_c governs nonlocal and dynamic effects. The term $2\eta\omega^2$ suppresses internal relaxation at high frequency, linking nonlinear softening with dispersion within a single model.

Within this physical picture, nonlinear softening is not empirically imposed, but naturally emerges from the interplay between coupling mechanisms and relaxation. At very small strains, $\mu_{\text{rel}} \approx \mu_{\text{rel}}^0$ and internal modes are strongly constrained, leading to a stiff response. As shear strain increases, $\mu_{\text{rel}}(\gamma)$ decreases, progressively enabling internal relaxation mechanisms associated with phenomena such as grain rearrangement, contact slip or crack opening to contribute more strongly to the effective macroscopic response.

Different microscopic mechanisms may lead to different evolutions of the coupling stiffness $\mu_e(\gamma)$. As a consequence, the present micromorphic formulation does not prescribe a unique softening mechanism, but provides a common mechanical framework within which different microstructural processes may be represented through their influence on the coupling between macroscopic and internal deformation. In other words, the model captures the effective response of internal mechanisms without requiring explicit specification of the underlying microstructure.

Although the inertial contribution is negligible under the experimental conditions investigated here, the formulation naturally extends to dynamic loading, where micro-inertia becomes significant and the model predicts the emergence of an internal resonance. In real materials this resonant behavior is expected to be regularized by dissipative mechanisms. The effective fitting parameters (A, B, C) are not purely empirical quantities, but can be directly related to the coupling stiffness, the relaxation stiffness and the micro-inertial contribution. In this sense, the fitting process provides a direct mechanical interpretation of the observed modulus reduction curves.

From this perspective, classical modulus reduction curves do not constitute independent constitutive laws, but rather represent low-frequency macroscopic manifestations of a more general micromorphic relaxation process.

4.2 Dynamic homogenization

Note that in general, the expressions derived for elastic parameters reveal a mathematical structure that can be written in the canonical form as follows

$$Q_{\text{macro}} = Q_{\text{micro}} \frac{\text{internal dynamics}}{\text{external coupling} + \text{internal dynamics}}, \quad (68)$$

where Q denotes a generic constitutive quantity. This structure reflects the underlying physics of the micromorphic framework. The macroscopic response emerges from a competition between internal microstructural dynamics and the stiffness of the surrounding medium. In particular, the frequency dependence enters through the term $L_c^2 \omega^2$, which governs the activation of internal deformation mechanisms.

Such a structure naturally arises in dynamic homogenization theories. In contrast to classical static homogenization, where the internal degrees of freedom are eliminated and only their average effect is retained, the micromorphic framework incorporates them explicitly, leading to frequency-dependent effective properties. A simple illustration of this mechanism is provided by mass-in-mass systems, for which the effective dynamic mass takes the form (Huang et al., 2009)

$$M_{\text{eff}}(\omega) = M + m \frac{\omega_0^2}{\omega_0^2 - \omega^2}, \quad (69)$$

where M is the macroscopic mass and m represents an internal oscillator with natural frequency ω_0 . The rational dependence on frequency arises from the interaction between macroscopic motion and internal dynamics (Srivastava, 2015).

Similarly, in elastodynamic homogenization theories, such as the framework introduced by Willis (Willis, 1980a,b, 1981, 1985), the effective constitutive relations become frequency dependent and include coupling terms between momentum and strain fields (Nassar et al., 2015). More recent developments show that both frequency and wavenumber dependence naturally emerge when microscale dynamics are retained in the upscaled description (Meng and Guzina, 2018; Blesgen and Neff, 2026). The micromorphic model provides a particularly transparent and physically interpretable dynamic homogenization. Rather than deriving effective parameters through asymptotic expansions, the model introduces internal degrees of freedom directly at the continuum level. Thus, the local field P should be understood as an effective continuum representation of the collective dynamics of the microstructure rather than as the motion of a specific microscopic constituent. As a result, dispersion and frequency-dependent behavior emerge naturally from the governing equations, without requiring explicit assumptions about the underlying microstructure.

4.3 Implications for distributed acoustic sensing (DAS) measurements

Distributed Acoustic Sensing (DAS) measures axial strain averaged over a finite gauge length L_g , as follows (e.g. Kennett, 2022)

$$\varepsilon^{\text{DAS}}(x) = \frac{1}{L_g} \int_{x-L_g/2}^{x+L_g/2} \partial_s u(s) ds. \quad (70)$$

In practice, the averaged strain rate is obtained by differentiating the ground velocity resolved along the cable at the ends of the gauge length L_g (Bakku, 2015). In numerical wave propagation highly complex and heterogeneous media, the gradient ∇u may exhibit strong spatial fluctuations or even do not mathematically exist $\nabla u \notin C^0(\Omega)$. To solve this problem, classical homogenization approaches are applied to regularize the gradient calculation (e.g. Capdeville and Sladen, 2024) and/or the Biot's theory of poroelasticity has been explored in realistic seismological scenarios of forward and inverse problems (e.g. Masson et al., 2006; Li and Schanz, 2011; Dupuy et al., 2016; Couples, 2019; Yang and Malcolm, 2021; Liu and Bohlen, 2023; Wang et al., 2024).

The presented micromorphic framework provides a dynamic counterpart of classical homogenization techniques used to model sub-wavelength heterogeneity effects (e.g. Capdeville and Sladen, 2024) and to the highly ill-posed inverse problem to determine Biot's elastic parameters (e.g. Wang et al., 2024). While on the one hand classical homogenization replaces micro-structure by an effective static medium, the micromorphic model retains additional internal degrees of freedom, which provide a natural regularization mechanism to the effective strain ($\nabla u - P$), and on the other hand, the model naturally accounts for both spatial averaging and frequency-dependent effects, allowing to properly reproduce the fundamental physics of wave propagation in poroelastic media. This offers a complete different physically grounded avenue to explore DAS measurements within highly heterogeneous media.

This is also relevant for near-surface geophysics, where the seismic record is also mainly dominated by surface waves. From a theoretical and numerical perspective, surface waves represent a complex subject because their interaction with highly heterogeneous media leads to very complex waveforms. As a consequence, their accurate modeling is crucial to properly image the interior of the Earth (Fichtner, 2010). Recent efforts have been taken in this direction in order to put surface waves in micro-continuum media into a rigorous mathematical foundation (Khan et al., 2022; Bulgariu et al., 2024; Apetrii et al., 2025).

5 Conclusions

The micromorphic framework provides a unified and physically consistent description of materials with microstructure. It is capable of simultaneously capturing nonlinear softening, frequency-dependent wave propagation, and scale effects within a single model. Unlike classical phenomenological approaches, the observed macroscopic behavior does not need to be prescribed a priori, but emerges naturally from the interaction between macroscopic deformation ∇u and internal degrees of freedom P .

We have shown that nonlinear softening can be understood as a consequence of the progressive reduction of the coupling between macroscopic deformation and internal local deformation. At small strains, the coupling stiffness is large and internal local deformation is constrained, leading to a stiff response. As strain increases, this constraint weakens, allowing internal relaxation mechanisms to become progressively active and reducing the effective macroscopic stiffness. In this sense, classical modulus reduction curves can be understood as macroscopic manifestations of an underlying relaxation process governed by internal kinematics.

From a theoretical perspective, the micromorphic framework can be interpreted as a kinematically enriched effective medium that extends classical static homogenization techniques. Rather than eliminating the microscale, it retains internal degrees of freedom that encode unresolved deformation mechanisms. Within this framework, the parameters do not correspond to distinct material phases, but to effective contributions controlling coupling, relaxation, and inertia.

Overall, the proposed framework provides a parsimonious yet powerful description of complex material behavior. With a small number of parameters, it captures nonlinear softening, dispersion, and scale-dependent effects without requiring explicit knowledge of the underlying microstructure. This makes the framework particularly well suited for realistic applications, where the detailed geometry of the medium is often unknown. More generally, the model offers a new perspective in which macroscopic material behavior is governed by the activation and suppression of internal relaxation mechanisms, rather than by the properties of distinct phases, providing a unified basis for interpreting experimental observations across scales.

From this perspective, classical modulus reduction laws are not viewed as independent constitutive relations, but as particular macroscopic limits of a more general micromorphic relaxation theory. This provides a direct mechanical interpretation of empirical softening laws while naturally extending them to include nonlocality, dispersion, and frequency-dependent effects within a unified constitutive framework.

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7 Data availability

The experimental data used for model validation were digitized from the published results of Dammala et al. (2019). No new experimental data were generated in this study. All equations required to reproduce the theoretical results are provided in the manuscript.

8 Conflict of interest

The authors have no conflicts of interest.

References

- Alavi, S. E., Ganghoffer, J.-F., Reda, H., and Sadighi, M. (2021). Construction of micromorphic continua by homogenization based on variational principles. *Journal of the Mechanics and Physics of Solids*, 153:104278.
- Apetrii, M., Bulgariu, E., Ghiba, I.-D., Khan, H., and Neff, P. (2025). Propagation of love waves in linear elastic isotropic Cosserat materials. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 481(2324):20250433.
- Bakku, S. K. (2015). *Fracture characterization from seismic measurements in a borehole*. PhD thesis, Massachusetts Institute of Technology.
- Barbagallo, G., Tallarico, D., d’Agostino, M. V., Aivaliotis, A., Neff, P., and Madeo, A. (2019). Relaxed micromorphic model of transient wave propagation in anisotropic band-gap metastructures. *International Journal of Solids and Structures*, 162:148–163.
- Beresnev, I. A. and Wen, K.-L. (1996). Nonlinear soil response—a reality? *Bulletin of the Seismological Society of America*, 86(6):1964–1978.
- Biswas, R. and Poh, L. H. (2017). A micromorphic computational homogenization framework for heterogeneous materials. *Journal of the Mechanics and Physics of Solids*, 102:187–208.
- Blesgen, T. and Neff, P. (2026). Global existence and uniqueness of weak solutions for a Willis-type model of elastodynamics. *Mechanics of Materials*, 213:105561.
- Bulgariu, E., Ghiba, I.-D., Khan, H., and Neff, P. (2024). Rayleigh waves in isotropic elastic materials with micro-voids. *International Journal of Solids and Structures*, 290:112661.
- Capdeville, Y., Guillot, L., and Marigo, J.-J. (2010). 2-D non-periodic homogenization to upscale elastic media for P–SV waves. *Geophysical Journal International*, 182(2):903–922.
- Capdeville, Y. and Métivier, L. (2018). Elastic full waveform inversion based on the homogenization method: theoretical framework and 2-D numerical illustrations. *Geophysical Journal International*, 213(2):1093–1112.
- Capdeville, Y. and Sladen, A. (2024). DAS sensitivity to heterogeneity scales much smaller than the minimum wavelength. *Seismica*, 3(1).
- Capdeville, Y., Stutzmann, E., Wang, N., and Montagner, J.-P. (2013). Residual homogenization for seismic forward and inverse problems in layered media. *Geophysical Journal International*, 194(1):470–487.
- Capdeville, Y., Zhao, M., and Cupillard, P. (2015). Fast Fourier homogenization for elastic wave propagation in complex media. *Wave Motion*, 54:170–186.
- Cosserat, E. and Cosserat, F. (1909). *Théorie des Corps Déformables*. Librairie Scientifique, A. Hermann et Fils, (english translation by D. Delphenich 2007, pdf available at [click HERE](#)), reprint 2009 by Hermann Librairie Scientifique, ISBN 978 27056 6920 1, Paris.
- Couples, G. D. (2019). Phenomenological understanding of poroelasticity via the micromechanics of a simple digital-rock model. *Geophysics*, 84(4):WA161–WA182.
- d’Agostino, M. V., Rizzi, G., Khan, H., Lewintan, P., Madeo, A., and Neff, P. (2022). The consistent coupling boundary condition for the classical micromorphic model: existence, uniqueness and interpretation of parameters. *Continuum Mechanics and Thermodynamics*, 34(6):1393–1431.
- Dammala, P. K., Kumar, S. S., Krishna, A. M., and Bhattacharya, S. (2019). Dynamic soil properties and liquefaction potential of northeast indian soil for non-linear effective stress analysis. *Bulletin of Earthquake Engineering*, 17(6):2899–2933.
- Darendeli, M. B. (2001). *Development of a new family of normalized modulus reduction and material damping*. Phd thesis, University of Texas, Austin, TX, USA.
- Demetriou, P., Voss, J., and Madeo, A. (2025). Effective interface forces to model boundary effects in a finite-size metamaterial through the reduced relaxed micromorphic model. *Mathematics and Mechanics of Solids*, page 10812865251358258.
- Dupuy, B., Asnaashari, A., Brossier, R., Garambois, S., Métivier, L., Ribodetti, A., and Virieux, J. (2016). A downscaling strategy from FWI to microscale reservoir properties from high-resolution images. *The Leading Edge*, 35(2):146–150.

- d'Agostino, M. V., Barbagallo, G., Ghiba, I.-D., Eidel, B., Neff, P., and Madeo, A. (2020). Effective description of anisotropic wave dispersion in mechanical band-gap metamaterials via the relaxed micromorphic model. *Journal of Elasticity*, 139(2):299–329.
- Eringen, A. C. and Claus Jr, W. D. (1970). A micromorphic approach to dislocation theory and its relation to several existing theories. In Simmons, J., de Wit, R., and Bullough, R., editors, *Fundamental Aspects of Dislocation Theory*, volume 1, page 1023–1040. Nat. Bur. Stand. (U.S.), Spec. Publ., Spec. Publ.
- Eringen, C. (1999). *Microcontinuum Field Theories I: Foundations and Solids*. Springer.
- Eringen, C. and Kafadar, C. (1976). *Polar Field Theories, Continuum Physics*, volume IV. Academic Press.
- Esfahani, R., Cotton, F., and Bonilla, L. F. (2024). Temporal variations of the ‘in-situ’ nonlinear behaviour of shallow sediments during the 2016 Kumamoto Earthquake sequence. *Geophysical Journal International*, 238(3):1626–1637.
- Fichtner, A. (2010). *Full Seismic Waveform Modelling and Inversion*. Springer.
- Forest, S. (2002). Homogenization methods and mechanics of generalized continua-part 2. *Theoretical and Applied Mechanics*, 28-29:113–144.
- Ganghoffer, J.-F., Wazne, A., and Reda, H. (2023). Frontiers in homogenization methods towards generalized continua for architected materials. *Mechanics Research Communications*, 130:104114.
- Gélis, C. and Bonilla, L. F. (2012). 2-DP-SV numerical study of soil–source interaction in a non-linear basin. *Geophysical Journal International*, 191(3):1374–1390.
- Grekova, E. (2012a). Linear reduced Cosserat medium with spherical tensor of inertia, where rotations are not observed in experiment. *Mechanics of Solids*, 47(5):538–543.
- Grekova, E. (2012b). Nonlinear isotropic elastic reduced Cosserat continuum as a possible model for geomedium and geomaterials. Spherical prestressed state in the semilinear material. *Journal of Seismology*, 16(4):695–707.
- Grekova, E. (2016). Plane waves in the linear elastic reduced Cosserat medium with a finite axially symmetric coupling between volumetric and rotational strains. *Mathematics and Mechanics of Solids*, 21(1):73–93.
- Grekova, E., Kulesh, M., and Herman, G. (2009). Waves in linear elastic media with microrotations, Part 2: Isotropic reduced Cosserat model. *Bulletin of the Seismological Society of America*, 99(2B):1423–1428.
- Guéguen, P., Bonilla, L. F., and Douglas, J. (2019). Comparison of soil nonlinearity (in situ stress–strain relation and G/G max reduction) observed in strong-motion databases and modeled in ground-motion prediction equations. *Bulletin of the Seismological Society of America*, 109(1):178–186.
- Hardin, B. O. (1965). The nature of damping in sands. *Journal of the Soil Mechanics and Foundations Division*, 91(1):63–97.
- Hardin, B. O. and Drnevich, V. P. (1972). Shear modulus and damping in soils: design equations and curves. *Journal of the Soil Mechanics and Foundations Division*, 98(7):667–692.
- Hartzell, S., Bonilla, L., and Williams, R. A. (2004). Prediction of nonlinear soil effects. *Bulletin of the Seismological Society of America*, 94(5):1609–1629.
- Hedjazian, N., Capdeville, Y., and Bodin, T. (2021). Multiscale seismic imaging with inverse homogenization. *Geophysical Journal International*, 226(1):676–691.
- Hermann, S., Billon, K., Collet, M., and Madeo, A. (2026). Unveiling the key role of interfaces in the design of finite-sized metamaterial structures. *European Journal of Mechanics-A/Solids*, page 106102.
- Hermann, S., Billon, K., Parlak, A.-M., Orlowsky, J., Collet, M., and Madeo, A. (2024). Design and experimental validation of a finite-size labyrinthine metamaterial for vibro-acoustics: enabling upscaling towards large-scale structures. *Philosophical Transactions A*, 382(2278):20230367.
- Huang, H., Sun, C., and Huang, G. (2009). On the negative effective mass density in acoustic metamaterials. *International Journal of Engineering Science*, 47(4):610–617.
- Hütter, G. (2017). Homogenization of a Cauchy continuum towards a micromorphic continuum. *Journal of the Mechanics and Physics of Solids*, 99:394–408.

- Hütter, G. (2019). On the micro-macro relation for the microdeformation in the homogenization towards micromorphic and micropolar continua. *Journal of the Mechanics and Physics of Solids*, 127:62–79.
- Ishihara, K. (1996). *Soil Behaviour in Earthquake Geotechnics*. Oxford University Press.
- Jeong, J. and Neff, P. (2010). Existence, uniqueness and stability in linear Cosserat elasticity for weakest curvature conditions. *Mathematics and Mechanics of Solids*, 15(1):78–95.
- Jeong, J., Ramézani, H., Münch, I., and Neff, P. (2009). A numerical study for linear isotropic Cosserat elasticity with conformally invariant curvature. *Zeitschrift für Angewandte Mathematik und Mechanik*, 89(7):552–569.
- Ji, C., Cabas, A., Bonilla, L. F., and Gelis, C. (2021). Effects of nonlinear soil behavior on Kappa (κ): Observations from the KiK-Net database. *Bulletin of the Seismological Society of America*, 111(4):2138–2157.
- Kennett, B. L. (2022). The seismic wavefield as seen by distributed acoustic sensing arrays: local, regional and teleseismic sources. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 478(2258).
- Khan, H., Ghiba, I.-D., Madeo, A., and Neff, P. (2022). Existence and uniqueness of Rayleigh waves in isotropic elastic Cosserat materials and algorithmic aspects. *Wave Motion*, 110:102898.
- Kulesh, M., Grekova, E., and Shardakov, I. (2009). The problem of surface wave propagation in a reduced Cosserat medium. *Acoustical Physics*, 55(2):218–226.
- Lai, S.-T., Schibuola, A., and Bonilla, L. F. (2025). Unveiling nonlinear site response through time-frequency analysis of earthquake records. *Geophysical Journal International*, 241(3):1601–1632.
- Lemaitre, J. (1992). *A Course on Damage Mechanics*. Springer.
- Li, P. and Schanz, M. (2011). Wave propagation in a 1-D partially saturated poroelastic column. *Geophysical journal international*, 184(3):1341–1353.
- Liu, T. and Bohlen, T. (2023). Time-domain poroelastic full-waveform inversion of shallow seismic data: methodology and sensitivity analysis. *Geophysical Journal International*, 232(3):1803–1820.
- Madeo, A., Corte, A. D., Giorgio, I., and Scerrato, D. (2017). Modeling and designing micro-and nano-structured metamaterials: towards the application of exotic behaviors. *Mathematics and Mechanics of Solids*, 22(4):873–884.
- Madeo, A., Neff, P., d’Agostino, M. V., and Barbagallo, G. (2016a). Complete band gaps including non-local effects occur only in the relaxed micromorphic model. *Comptes Rendus Mécanique*, 344(11):784 – 796.
- Madeo, A., Neff, P., Ghiba, I.-D., Placidi, L., and Rosi, G. (2015). Wave propagation in relaxed micromorphic continua: modeling metamaterials with frequency band-gaps. *Continuum Mechanics and Thermodynamics*, 27(4-5):551–570.
- Madeo, A., Neff, P., Ghiba, I.-D., and Rosi, G. (2016b). Reflection and transmission of elastic waves in non-local band-gap metamaterials: A comprehensive study via the relaxed micromorphic model. *Journal of the Mechanics and Physics of Solids*, 95:441 – 479.
- Masson, Y. J., Pride, S. R., and Nihei, K. (2006). Finite difference modeling of Biot’s poroelastic equations at seismic frequencies. *Journal of Geophysical Research: Solid Earth*, 111(B10).
- Meng, S. and Guzina, B. B. (2018). On the dynamic homogenization of periodic media: Willis’ approach versus two-scale paradigm. *Proceedings of the Royal Society A*, 474(2213):20170638.
- Mindlin, R. D. (1964). Micro-structure in linear elasticity. *Archive for Rational Mechanics and Analysis*, 16(1):51–78.
- Moczo, P., Kristek, J., and Gális, M. (2014). *The Finite-Difference Modelling of Earthquake Motions: Waves and Ruptures*. Cambridge University Press.
- Nagahama, H. and Teisseyre, R. (2000a). Micromorphic continuum and fractal fracturing in the lithosphere. *Pure and Applied Geophysics*, 157(4):559–574.
- Nagahama, H. and Teisseyre, R. (2000b). Micromorphic continuum and fractal properties of faults and earthquakes. *Earthquake Thermodynamics and Phase Transformations in the Earth’s Interior*, pages 425–440.
- Nassar, H., He, Q.-C., and Auffray, N. (2015). Willis elastodynamic homogenization theory revisited for periodic media. *Journal of the Mechanics and Physics of Solids*, 77:158–178.

- Neff, P., Eidel, B., d'Agostino, M. V., and Madeo, A. (2020). Identification of scale-independent material parameters in the relaxed micromorphic model through model-adapted first order homogenization. *Journal of Elasticity*, 139(2):269–298.
- Neff, P., Ghiba, I.-D., Madeo, A., Placidi, L., and Rosi, G. (2014). A unifying perspective: the relaxed linear micromorphic continuum. *Continuum Mechanics and Thermodynamics*, 26(5):639–681.
- Neff, P. and Jeong, J. (2009). A new paradigm: the linear isotropic Cosserat model with conformally invariant curvature energy. *Zeitschrift für Angewandte Mathematik und Mechanik*, 89(2):107–122.
- Nowacki, W. (1986). *Theory of Micropolar Elasticity*. Pergamon Press.
- Pyke, R. M. (1979). Nonlinear soil models for irregular cyclic loadings. *Journal of the Geotechnical Engineering Division*, 105(6):715–726.
- Ramirez, L. A. P., Erel-Demore, F., Rizzi, G., Voss, J., and Madeo, A. (2024). Effective surface forces and non-coherent interfaces within the reduced relaxed micromorphic modeling of finite-size mechanical metamaterials. *Journal of the Mechanics and Physics of Solids*, 186:105558.
- Rizzi, G., d'Agostino, M. V., Voss, J., Bernardini, D., Neff, P., and Madeo, A. (2024). From frequency-dependent models to frequency-independent enriched continua for mechanical metamaterials. *European Journal of Mechanics-A/Solids*, 106:105269.
- Rokoš, O., Ameen, M., Peerlings, R., and Geers, M. (2020a). Extended micromorphic computational homogenization for mechanical metamaterials exhibiting multiple geometric pattern transformations. *Extreme Mechanics Letters*, 37:100708.
- Rokoš, O., Ameen, M. M., Peerlings, R. H., and Geers, M. G. (2019). Micromorphic computational homogenization for mechanical metamaterials with patterning fluctuation fields. *Journal of the Mechanics and Physics of Solids*, 123:119–137.
- Rokoš, O., Zeman, J., Doškář, M., and Krysl, P. (2020b). Reduced integration schemes in micromorphic computational homogenization of elastomeric mechanical metamaterials. *Advanced Modeling and Simulation in Engineering Sciences*, 7(1):19.
- Sarhil, M., Scheunemann, L., Lewintan, P., Schröder, J., and Neff, P. (2024). A computational approach to identify the material parameters of the relaxed micromorphic model. *Computer Methods in Applied Mechanics and Engineering*, 425:116944.
- Sarhil, M., Scheunemann, L., Schröder, J., and Neff, P. (2023a). On the identification of material parameters in the relaxed micromorphic continuum. *Proceedings in Applied Mathematics and Mechanics*, 23(3):e202300056.
- Sarhil, M., Scheunemann, L., Schröder, J., and Neff, P. (2023b). Size-effects of metamaterial beams subjected to pure bending: on boundary conditions and parameter identification in the relaxed micromorphic model. *Computational Mechanics*, 72(5):1091–1113.
- Shimbo, M. (1975). A geometrical formulation of asymmetric features in plasticity. *Bulletin of the Faculty of Engineering, Hokkaido University*, 77(77):155–159.
- Sky, A., Muench, I., Rizzi, G., and Neff, P. (2024). Higher order Bernstein–Bézier and Nédélec finite elements for the relaxed micromorphic model. *Journal of Computational and Applied Mathematics*, 438:115568.
- Sky, A., Neunteufel, M., Lewintan, P., Gourgietis, P., Zilian, A., and Neff, P. (2025). Novel H^1 dev (Curl)-conforming elements on regular triangulations and Clough–Tocher splits for the planar relaxed micromorphic model. *Computational Mechanics*, 76(2):517–539.
- Slawinski, M. (2015). *Waves and Rays in Elastic Continua*. World Scientific.
- Srivastava, A. (2015). Elastic metamaterials and dynamic homogenization: a review. *International Journal of Smart and Nano Materials*, 6(1):41–64.
- Teisseyre, R. (1973). Earthquake processes in a micromorphic continuum. *Pure and Applied Geophysics*, 102(1):15–28.
- Teisseyre, R. (1974). Symmetric Micromorphic Continuum: Wave Propagation, Point Source Solutions and Some Applications to Earthquake Processes. In Thoft-Christensen, P., editor, *Continuum Mechanics Aspects of Geodynamics and Rock Fracture Mechanics: proceedings of the NATO advanced study institute held in Reykjavik, Iceland, 11-20 August, 1974*, pages 201–244. Springer.
- Teisseyre, R. (2008). Friction and Fracture Induced Anisotropy: Asymmetric Stresses. In Teisseyre, R., Nagahama, H., and Majewski, E., editors, *Physics of Asymmetric Continuum: Extreme and Fracture Processes*, pages 163–169. Springer.
- Teisseyre, R. (2011). Why rotation seismology: confrontation between classic and asymmetric theories. *Bulletin of the Seismological Society of America*, 101(4):1683–1691.
- Teisseyre, R. and Górski, M. (2009). Fundamental deformations in asymmetric continuum. *Bulletin of the Seismological Society of America*, 99(2B):1132–1136.

- Teisseyre, R., Gorski, M., and Teisseyre, K. (2008). Fracture Processes: Spin and Twist-Shear Coincidence. In Teisseyre, R., Nagahama, H., and Majewski, E., editors, *Physics of Asymmetric Continuum: Extreme and Fracture Processes*, pages 111–122. Springer.
- Teisseyre, R. and Górski, M. (2007). Fundamental Deformations in Asymmetric Continuum: Motions and Fracturing. In Lee, W., Çelebi, M., Todorovska, M., and Diggles, M., editors, *Rotational Seismology and Engineering Applications—Online Proceedings for the First International Workshop*. U.S. Geological Survey.
- Teisseyre, R., Takeo, M., and Majewski, E. (2006). *Earthquake Source Asymmetry, Structural Media and Rotation Effects*. Springer.
- Trinh, D. K., Janicke, R., Auffray, N., Diebels, S., and Forest, S. (2012). Evaluation of generalized continuum substitution models for heterogeneous materials. *International Journal for Multiscale Computational Engineering*, 10(6).
- Voss, J., Rizzi, G., Demetriou, P., Neff, P., and Madeo, A. (2023). Remarks on wave propagation in an acoustic metamaterial modeled as a relaxed micromorphic continuum. *Proceedings in Applied Mathematics and Mechanics*, 23(1):e202200182.
- Wang, H., Liu, Y., Liu, L., Yan, T., and Zhang, Z. (2024). Petrophysical parameter estimation using Biot-poroelastic full-waveform inversion. *Geophysical Journal International*, 237(3):1639–1654.
- Willis, J. (1980a). Polarization approach to the scattering of elastic waves—I. Scattering by a single inclusion. *Journal of the Mechanics and Physics of Solids*, 28(5-6):287–305.
- Willis, J. (1980b). A polarization approach to the scattering of elastic waves—II. Multiple scattering from inclusions. *Journal of the Mechanics and Physics of Solids*, 28(5-6):307–327.
- Willis, J. R. (1981). Variational principles for dynamic problems for inhomogeneous elastic media. *Wave Motion*, 3(1):1–11.
- Willis, J. R. (1985). The nonlocal influence of density variations in a composite. *International Journal of Solids and Structures*, 21(7):805–817.
- Yang, Q. and Malcolm, A. (2021). Frequency domain full-waveform inversion in a fluid-saturated poroelastic medium. *Geophysical Journal International*, 225(1):68–84.
- Zhang, J., Andrus, R. D., and Juang, C. H. (2005). Normalized shear modulus and material damping ratio relationships. *Journal of Geotechnical and Geoenvironmental Engineering*, 131(4):453–464.
- Zhi, J., Poh, L. H., Tay, T.-E., and Tan, V. B. C. (2022). Direct FE2 modeling of heterogeneous materials with a micromorphic computational homogenization framework. *Computer Methods in Applied Mechanics and Engineering*, 393:114837.